empirical references

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This document contains the sources and references for the empirical data we used in *Extending Bayesian Elo-rating to quantify the steepness of dominance hierarchies* (https://doi.org/10.1101/2022.01.28.478016). It was built using `EloSteepness.data` (version 0.9.6).

The actual matrices are in:

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library(EloSteepness.data)
empirical_matlist
```

Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
addison1982_dom_2	*Capra hircus*	goat	9	106	0.28	Addison & Baker (1982)
alados1992_dom_1	*Nanger dama*	dama gazelle	7	298	0.00	Alados & Escós (1992)
alados1992_dom_2	*Nanger dama*	dama gazelle	9	324	0.03	Alados & Escós (1992)
alados1992_dom_3	*Nanger dama*	dama gazelle	7	270	0.05	Alados & Escós (1992)
alados1992_dom_5	*Gazella cuvieri*	Cuvier's gazelle	5	658	0.00	Alados & Escós (1992)
alados1992_dom_6	*Gazella cuvieri*	Cuvier's gazelle	8	666	0.00	Alados & Escós (1992)
alados1992_dom_7	*Gazella cuvieri*	Cuvier's gazelle	7	604	0.00	Alados & Escós (1992)
allee1954_dom_1	*Mustelus canis*	smooth dogfish	10	267	0.38	Allee & Dickinson (1954); Shizuka & McDonald (2012)
appleby1983_dom_1	*Cervus elaphus*	red deer	7	83	0.38	Appleby (1983)
archie2006_dom_02	*Loxodonta africana*	African elephant	6	48	0.07	Archie et al. (2006)
archie2006_dom_03	*Loxodonta africana*	African elephant	5	25	0.10	Archie et al. (2006)
archie2006_dom_05	*Loxodonta africana*	African elephant	10	64	0.44	Archie et al. (2006)
archie2006_dom_06	*Loxodonta africana*	African elephant	6	39	0.13	Archie et al. (2006)
archie2006_dom_08	*Loxodonta africana*	African elephant	7	26	0.43	Archie et al. (2006)
archie2006_dom_11	*Loxodonta africana*	African elephant	8	62	0.21	Archie et al. (2006)
archie2006_dom_12	*Loxodonta africana*	African elephant	5	24	0.00	Archie et al. (2006)
archie2006_dom_13	*Loxodonta africana*	African elephant	5	33	0.00	Archie et al. (2006)
archie2006_dom_15	*Loxodonta africana*	African elephant	5	24	0.30	Archie et al. (2006)
archie2006_dom_16	*Loxodonta africana*	African elephant	5	35	0.00	Archie et al. (2006)
archie2006_dom_18	*Loxodonta africana*	African elephant	6	33	0.07	Archie et al. (2006)
arlet2009_dom_4	*Lophocebus albigena*	gray-cheeked mangabey	5	73	0.00	Arlet & Isbell (2009)
arlet2009_dom_5	*Lophocebus albigena*	gray-cheeked mangabey	9	257	0.00	Arlet & Isbell (2009)
arlet2011_dom_3	*Lophocebus albigena*	gray-cheeked mangabey	5	63	0.00	Arlet et al. (2011)
athman2010_dom_1	*Macaca radiata*	bonnet macaque	7	80	0.00	Athman (2010)
balasubramaniam2014_dom_1	*Macaca tonkeana*	Tonkean macaque	9	174	0.14	Balasubramaniam (2014); Demaria & Thierry (2001)
balasubramaniam2014_dom_2	*Macaca nigra*	crested macaque	6	516	0.00	Balasubramaniam (2014)
banks1956_dom_1	*Gallus gallus*	red jungle fowl	9	1097	0.00	Banks (1956)
banks1956_dom_2	*Gallus gallus*	red jungle fowl	9	970	0.00	Banks (1956)
banks1956_dom_3	*Gallus gallus*	red jungle fowl	10	1164	0.00	Banks (1956)
banks1956_dom_4	*Gallus gallus*	red jungle fowl	10	1200	0.00	Banks (1956)
banks1956_dom_5	*Gallus gallus*	red jungle fowl	9	1297	0.00	Banks (1956)
banks1956_dom_6	*Gallus gallus*	red jungle fowl	9	1093	0.00	Banks (1956)
banks1956_dom_7	*Gallus gallus*	red jungle fowl	10	1023	0.00	Banks (1956)
banks1956_dom_8	*Gallus gallus*	red jungle fowl	10	1351	0.00	Banks (1956)

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data set	species	common name	group size	interactions	sparseness	reference
barrette1986_dom_1	*Rangifer tarandus*	woodland caribou	20	823	0.06	Barrette & Vandal (1986)
bartlett1971_dom_1	*Macaca mulatta*	rhesus macaque	13	85	0.23	Bartlett & Meier (1971)
bartlett1971_dom_2	*Macaca mulatta*	rhesus macaque	13	138	0.18	Bartlett & Meier (1971)
barton1993_dom_1	*Papio anubis*	olive baboon	26	511	0.34	Barton & Whiten (1993)
beilharz1982_dom_1	*Bos taurus*	dairy cow	11	59	0.44	Beilharz & Zeeb (1982)
beilharz1982_dom_2	*Bos taurus*	dairy cow	9	60	0.36	Beilharz & Zeeb (1982)
bell1978_dom_1	*Nauphoeta cinerea*	cockroach	5	101	0.00	Bell & Gorton (1978)
bell1978_dom_2	*Nauphoeta cinerea*	cockroach	5	92	0.00	Bell & Gorton (1978)
bell1978_dom_3	*Nauphoeta cinerea*	cockroach	5	123	0.10	Bell & Gorton (1978)
bell1978_dom_4	*Nauphoeta cinerea*	cockroach	5	149	0.00	Bell & Gorton (1978)
bell1978_dom_5	*Nauphoeta cinerea*	cockroach	5	161	0.00	Bell & Gorton (1978)
bell1978_dom_6	*Nauphoeta cinerea*	cockroach	5	75	0.10	Bell & Gorton (1978)
bell1978_dom_7	*Nauphoeta cinerea*	cockroach	5	78	0.20	Bell & Gorton (1978)
bell1978_dom_8	*Nauphoeta cinerea*	cockroach	5	144	0.00	Bell & Gorton (1978)
bell1978_dom_9	*Nauphoeta cinerea*	cockroach	5	62	0.30	Bell & Gorton (1978)
bennett1939_dom_1	*Streptopelia risoria*	ring dove	5	58	0.10	M. A. Bennett (1939)
bennett1939_dom_2	*Streptopelia risoria*	ring dove	9	628	0.00	M. A. Bennett (1939)
bennett1939_dom_3	*Streptopelia risoria*	ring dove	5	492	0.00	M. A. Bennett (1939)
bennett1939_dom_4	*Streptopelia risoria*	ring dove	5	795	0.00	M. A. Bennett (1939)
bennett1939_dom_5	*Streptopelia risoria*	ring dove	5	1013	0.00	M. A. Bennett (1939)
bennett1939_dom_6	*Streptopelia risoria*	ring dove	5	560	0.00	M. A. Bennett (1939)
bennett1986_dom_1	*Ovis canadensis*	bighorn sheep	6	89	0.00	B. Bennett (1986)
bercovitch1993_dom_1	*Macaca mulatta*	rhesus macaque	9	77	0.25	Bercovitch (1993)
bergerud1974_dom_1	*Rangifer tarandus*	caribou	9	46	0.42	Bergerud (1974)
bergstrom2010_dom_1	*Cebus capucinus*	white-faced capuchin	5	24	0.00	Bergstrom & Fedigan (2010)
bergstrom2010_dom_2	*Cebus capucinus*	white-faced capuchin	7	117	0.05	Bergstrom & Fedigan (2010)
bergstrom2010_dom_3	*Cebus capucinus*	white-faced capuchin	10	155	0.20	Bergstrom & Fedigan (2010)
berman2004_dom_1	*Macaca thibetana*	Tibetan macaque	22	858	0.21	Berman et al. (2004)
berman2004_dom_2	*Macaca thibetana*	Tibetan macaque	22	871	0.28	Berman et al. (2004)
berman2004_dom_3	*Macaca thibetana*	Tibetan macaque	18	1521	0.12	Berman et al. (2004)
berryman1978_dom_1	*Cavia porcellus*	guinea pig	8	170	0.21	Berryman (1978)
birky2002_dom_01	*Macaca cyclopis*	Formosan macaque	6	109	0.00	Birky (2002)
birky2002_dom_02	*Macaca cyclopis*	Formosan macaque	6	61	0.07	Birky (2002)
birky2002_dom_11	*Macaca cyclopis*	Formosan macaque	5	107	0.00	Birky (2002)
birky2002_dom_12	*Macaca cyclopis*	Formosan macaque	9	256	0.03	Birky (2002)
blatrix2004_dom_1	*Temnothorax americanus*	ant	16	629	0.41	Blatrix & Herbers (2004)
blatrix2004_dom_2	*Temnothorax americanus*	ant	8	46	0.43	Blatrix & Herbers (2004)
blatrix2004_dom_5	*Temnothorax americanus*	ant	6	53	0.33	Blatrix & Herbers (2004)
bonanni2007_dom_1	*Felis catus*	domestic cat	13	360	0.28	Bonanni et al. (2007)
bonanni2007_dom_2	*Felis catus*	domestic cat	14	565	0.37	Bonanni et al. (2007)
bonanni2007_dom_3	*Felis catus*	domestic cat	14	307	0.44	Bonanni et al. (2007)
borries1991_dom_1	*Semnopithecus entellus*	Hanuman langur	11	149	0.09	Borries et al. (1991)
borries1991_dom_2	*Semnopithecus entellus*	Hanuman langur	11	166	0.20	Borries et al. (1991)
borries1991_dom_3	*Semnopithecus entellus*	Hanuman langur	11	171	0.09	Borries et al. (1991)
borries1991_dom_4	*Semnopithecus entellus*	Hanuman langur	13	437	0.06	Borries et al. (1991)
borries1991_dom_5	*Semnopithecus entellus*	Hanuman langur	13	520	0.08	Borries et al. (1991)
borries1991_dom_6	*Semnopithecus entellus*	Hanuman langur	13	420	0.05	Borries et al. (1991)
borries1991_dom_7	*Semnopithecus entellus*	Hanuman langur	12	296	0.08	Borries et al. (1991)
borries1991_dom_8	*Semnopithecus entellus*	Hanuman langur	12	521	0.00	Borries et al. (1991)
borries1991_dom_9	*Semnopithecus entellus*	Hanuman langur	12	260	0.06	Borries et al. (1991)
bromley1991_dom_1	*Antilocapra americana*	pronghorn	8	130	0.14	Bromley (1991)
brown1963_dom_01	*Cyanocitta stelleri*	Steller jay	18	600	0.29	Brown (1963)
brown1963_dom_02	*Cyanocitta stelleri*	Steller jay	5	64	0.10	Brown (1963)
brown1963_dom_03	*Cyanocitta stelleri*	Steller jay	5	48	0.00	Brown (1963)
brown1963_dom_04	*Cyanocitta stelleri*	Steller jay	5	48	0.20	Brown (1963)
brown1963_dom_06	*Cyanocitta stelleri*	Steller jay	5	39	0.20	Brown (1963)
brown1963_dom_07	*Cyanocitta stelleri*	Steller jay	5	63	0.00	Brown (1963)

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data set	species	common name	group size	interactions	sparseness	reference
brown1963_dom_08	*Cyanocitta stelleri*	Steller jay	5	25	0.00	Brown (1963)
brown1963_dom_11	*Cyanocitta stelleri*	Steller jay	20	453	0.45	Brown (1963)
bruinzeel2006_dom_1	*Haematopus ostralegus*	oystercatcher	6	183	0.27	Bruinzeel et al. (2006)
buck1998_dom_1	*Cercocebus atys*	sooty mangabey	5	104	0.00	Buck (1998)
buck1998_dom_2	*Cercocebus atys*	sooty mangabey	6	150	0.00	Buck (1998)
butovskaya1996_dom_1	*Macaca fascicularis*	longtailed macaque	8	322	0.04	Butovskaya et al. (1996)
butovskaya1996_dom_2	*Macaca fascicularis*	longtailed macaque	10	605	0.09	Butovskaya et al. (1996)
byrne1983_dom_1	*Cercopithecus diana*	Diana monkey	5	404	0.00	Byrne et al. (1983)
cafazzo2010_dom_1	*Canis familiaris*	dog	27	1815	0.23	Cafazzo et al. (2010)
canali1986_dom_1	*Bos taurus*	domestic cattle	9	746	0.17	Canali et al. (1986)
carlisle1986_dom_03	*Argya squamiceps*	Arabian babbler	8	87	0.36	Carlisle & Zahavi (1986)
carlisle1986_dom_06	*Argya squamiceps*	Arabian babbler	9	80	0.31	Carlisle & Zahavi (1986)
carlisle1986_dom_07	*Argya squamiceps*	Arabian babbler	9	86	0.25	Carlisle & Zahavi (1986)
carlisle1986_dom_11	*Argya squamiceps*	Arabian babbler	10	57	0.47	Carlisle & Zahavi (1986)
carlisle1986_dom_14	*Argya squamiceps*	Arabian babbler	5	105	0.00	Carlisle & Zahavi (1986)
carlisle1986_dom_15	*Argya squamiceps*	Arabian babbler	6	35	0.20	Carlisle & Zahavi (1986)
carlisle1986_dom_16	*Argya squamiceps*	Arabian babbler	8	137	0.11	Carlisle & Zahavi (1986)
carlisle1986_dom_18	*Argya squamiceps*	Arabian babbler	5	43	0.20	Carlisle & Zahavi (1986)
carlisle1986_dom_19	*Argya squamiceps*	Arabian babbler	6	18	0.40	Carlisle & Zahavi (1986)
castles1997_dom_1	*Papio anubis*	olive baboon	25	1080	0.16	Castles (1997)
chacko2014_dom_1	*Macaca radiata*	bonnet macaque	11	354	0.04	Chacko (2014)
chacko2014_dom_2	*Macaca radiata*	bonnet macaque	11	511	0.07	Chacko (2014)
chancellor2009_dom_1	*Lophocebus albigena*	gray-cheeked mangabey	7	115	0.14	Chancellor & Isbell (2009)
chancellor2009_dom_2	*Lophocebus albigena*	gray-cheeked mangabey	5	65	0.00	Chancellor & Isbell (2009)
chancellor2009_dom_3	*Lophocebus albigena*	gray-cheeked mangabey	5	98	0.00	Chancellor & Isbell (2009)
chancellor2009_dom_5	*Lophocebus albigena*	gray-cheeked mangabey	5	85	0.00	Chancellor & Isbell (2009)
chapais1985_dom_1	*Macaca fuscata*	Japanese macaque	10	97	0.40	Chapais (1985)
cheney1977_dom_1	*Papio ursinus*	chacma baboon	12	617	0.00	Cheney (1977)
cheney1977_dom_2	*Papio ursinus*	chacma baboon	12	395	0.01	Cheney (1977)
cheney1981_dom_1		vervet monkey	8	989	0.00	Cheney et al. (1981)
cheney1981_dom_2		vervet monkey	7	644	0.00	Cheney et al. (1981)
cheney1981_dom_3		vervet monkey	8	766	0.00	Cheney et al. (1981)
chism1997_dom_1	*Erythrocebus patas*	patas monkey	5	26	0.40	Chism & Rogers (1997)
clark1974_dom_1	*Saimiri sciureus*	squirrel monkey	10	45	0.00	Clark & Dillon (1974)
clutton-brock1976_dom_1	*Equus caballus*	highland pony	17	761	0.05	Clutton-Brock et al. (1976)
coffin1979_dom_1		vervet	8	928	0.00	Coffin (1979)
coffin1979_dom_2		vervet	8	486	0.04	Coffin (1979)
collias1950_dom_1	*Odocoileus virginianus*	white-tailed deer	9	472	0.00	Collias (1950); Shizuka & McDonald (2012)
collias1950_dom_2	*Tragelaphus angasii*	nyala	7	110	0.33	Collias (1950); Shizuka & McDonald (2012)
collias1951_dom_2	*Phasianus colchicus*	ring-necked pheasant	8	48	0.46	Collias & Taber (1951)
collias1951_dom_3	*Phasianus colchicus*	ring-necked pheasant	14	136	0.44	Collias & Taber (1951)
cooper1999_dom_1	*Macaca assamensis*	Assamese macaque	41	4767	0.19	Cooper (1999)
cords2000_dom_5	*Cercopithecus mitis*	blue monkey	19	356	0.28	Cords (2000)
cords2002_dom_1	*Cercopithecus mitis*	blue monkey	17	492	0.14	Cords (2002)
cote2000_dom_1	*Oreamnos americanus*	mountain goat	38	760	0.41	Côté (2000)
coutlee1967_dom_1	*Spinus tristis*	American goldfinch	5	567	0.00	Coutlee (1967)
cui2014_dom_1	*Rhinopithecus bieti*	black-and-white snub-nosed monkey	7	717	0.00	Cui et al. (2014)
cui2014_dom_2	*Rhinopithecus bieti*	black-and-white snub-nosed monkey	7	300	0.00	Cui et al. (2014)
cui2014_dom_3	*Rhinopithecus bieti*	black-and-white snub-nosed monkey	14	7825	0.01	Cui et al. (2014)
david2003_dom_1	*Galaxias argenteus*	giant kokopu	7	35	0.43	David & Stoffels (2003)
dawson1991_dom_1	*Parabuteo unicinctus*	Harris' hawk	5	96	0.10	Dawson & Mannan (1991)
deng1987_dom_1	*Macaca thibetana*	Tibetan macaque	24	988	0.25	Deng & Zhao (1987)
devries1993_dom_1	*Equus caballus*	horse	9	155	0.14	de Vries et al. (1993)
devries1995_dom_1	*Pongo pygmaeus*	orang utan	7	65	0.38	Utami and Wich in de Vries (1995)
devries2006_dom_1	*Pan paniscus*	bonobo	7	456	0.14	JMG Stevens in: de Vries et al. (2006)
dewaal1977_dom_1	*Macaca fascicularis*	longtailed macaque	14	1231	0.06	de Waal (1977)

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data set	species	common name	group size	interactions	sparseness	reference
dewaal1977_dom_2	*Macaca fascicularis*	longtailed macaque	17	1084	0.12	de Waal (1977)
dewaal1985_dom_1	*Macaca mulatta*	rhesus macaque	24	1275	0.11	de Waal & Luttrell (1985)
diniz2019_dom_1	*Cacicus haemorrhous*	red-rumped cacique	10	224	0.13	Diniz et al. (2019)
dixson1977_dom_1	*Miopithecus talapoin*	talapoin monkey	8	1124	0.07	Dixson & Herbert (1977)
dixson1977_dom_2	*Miopithecus talapoin*	talapoin monkey	8	1168	0.07	Dixson & Herbert (1977)
dixson1977_dom_3	*Miopithecus talapoin*	talapoin monkey	6	910	0.07	Dixson & Herbert (1977)
duboscq2013_dom_1	*Macaca nigra*	crested macaque	15	561	0.04	Duboscq et al. (2013)
duboscq2013_dom_2	*Macaca nigra*	crested macaque	15	210	0.23	Duboscq et al. (2013)
duboscq2013_dom_3	*Macaca nigra*	crested macaque	15	407	0.10	Duboscq et al. (2013)
duboscq2013_dom_4	*Macaca nigra*	crested macaque	21	283	0.36	Duboscq et al. (2013)
duboscq2013_dom_5	*Macaca nigra*	crested macaque	21	732	0.11	Duboscq et al. (2013)
duboscq2013_dom_6	*Macaca nigra*	crested macaque	21	711	0.12	Duboscq et al. (2013)
dunford1970_dom_1	*Tamias striatus*	eastern chipmunk	11	316	0.44	Dunford (1970)
easley1989_dom_1	*Papio anubis*	anubis baboon	8	160	0.07	Easley et al. (1989)
easley1989_dom_2	*Papio anubis*	anubis baboon	8	253	0.04	Easley et al. (1989)
eckert1987_dom_1	*Agelaius phoeniceus*	red-winged blackbird	8	973	0.00	Eckert & Weatherhead (1987)
eckert1987_dom_2	*Agelaius phoeniceus*	red-winged blackbird	10	2760	0.00	Eckert & Weatherhead (1987)
eckert1987_dom_3	*Agelaius phoeniceus*	red-winged blackbird	10	3423	0.00	Eckert & Weatherhead (1987)
ellard1989_dom_1	*Equus caballus*	horse	12	626	0.00	Ellard & Crowell-Davis (1989)
ellsworth1999_dom_6	*Megascops kennicottii*	western screech-owl	5	114	0.00	Ellsworth & Belthoff (1999)
ellsworth1999_dom_7	*Megascops kennicottii*	western screech-owl	5	31	0.10	Ellsworth & Belthoff (1999)
ellsworth1999_dom_8	*Megascops kennicottii*	western screech-owl	5	58	0.00	Ellsworth & Belthoff (1999)
erlinge1977_dom_1	*Mustela erminea*	stout	6	30	0.27	Erlinge (1977)
espmark1974_dom_1	*Capreolus capreolus*	roe deer	6	473	0.00	Espmark (1974)
espmark1974_dom_2	*Capreolus capreolus*	roe deer	6	399	0.00	Espmark (1974)
espmark1974_dom_3	*Capreolus capreolus*	roe deer	5	39	0.10	Espmark (1974)
espmark1974_dom_4	*Capreolus capreolus*	roe deer	11	145	0.33	Espmark (1974)
fairbanks1994_dom_1	*Antilocapra americana*	pronghorn	9	44	0.36	Fairbanks (1994)
fairbanks1994_dom_2	*Antilocapra americana*	pronghorn	12	123	0.38	Fairbanks (1994)
fairbanks1994_dom_3	*Antilocapra americana*	pronghorn	9	167	0.17	Fairbanks (1994)
fairbanks1994_dom_5	*Antilocapra americana*	pronghorn	11	157	0.11	Fairbanks (1994)
farentinos1972_dom_1	*Sciurus aberti*	tassel-eared squirrel	6	181	0.27	Farentinos (1972)
farentinos1972_dom_2	*Sciurus aberti*	tassel-eared squirrel	8	245	0.29	Farentinos (1972)
farentinos1972_dom_4	*Sciurus aberti*	tassel-eared squirrel	11	324	0.44	Farentinos (1972)
farentinos1972_dom_5	*Sciurus aberti*	tassel-eared squirrel	6	431	0.00	Farentinos (1972)
farentinos1972_dom_6	*Sciurus aberti*	tassel-eared squirrel	10	316	0.44	Farentinos (1972)
feh1998_dom_2	*Equus przewalskii*	Przewalski horse	5	63	0.00	Feh (1988)
fortunato2009_dom_1	*Pan paniscus*	bonobo	9	376	0.06	Fortunato (2009)
foster1981_dom_2	*Chiroxiphia caudata*	swallow-tailed manakin	5	125	0.30	Foster (1981)
freeman2013a_dom_1	*Chlorocebus pygerythrus*	vervet	14	439	0.15	Freeman (2012)
freeman2013a_dom_2	*Chlorocebus pygerythrus*	vervet	9	268	0.17	Freeman (2012)
fugle1984_dom_1	*Zonotrichia leucophrys*	white-crowned sparrow	8	446	0.00	Fugle et al. (1984)
fugle1984_dom_2	*Zonotrichia leucophrys*	white-crowned sparrow	8	418	0.00	Fugle et al. (1984)
fugle1984_dom_3	*Zonotrichia leucophrys*	white-crowned sparrow	8	157	0.04	Fugle et al. (1984)
fugle1984_dom_4	*Zonotrichia leucophrys*	white-crowned sparrow	9	547	0.00	Fugle et al. (1984)
fugle1984_dom_5	*Zonotrichia leucophrys*	white-crowned sparrow	10	449	0.00	Fugle et al. (1984)
fugle1984_dom_6	*Zonotrichia leucophrys*	white-crowned sparrow	10	434	0.00	Fugle et al. (1984)
fugle1984_dom_7	*Zonotrichia leucophrys*	white-crowned sparrow	10	514	0.00	Fugle et al. (1984)
fugle1984_dom_8	*Zonotrichia leucophrys*	white-crowned sparrow	11	472	0.00	Fugle et al. (1984)
funkhouser2018_dom_1	*Pan troglodytes*	chimpanzee	7	55	0.19	Funkhouser et al. (2018)
furuichi1997_dom_1	*Pan paniscus*	bonobo	9	75	0.33	Furuichi (1997)
gemmill2007_dom_1	*Lemur catta*	ringtailed lemur	6	320	0.00	Gemmill (2007)
gemmill2007_dom_2	*Lemur catta*	ringtailed lemur	6	245	0.00	Gemmill (2007)
graf1985_dom_1	*Lepus americanus*	snowshoe hare	5	242	0.00	Graf (1985)
greenberg-cohen1994_dom_1	*Capra nubiana*	Nubian ibex	13	414	0.05	Greenberg-Cohen et al. (1994)
grunau2001_dom_1	*Colobus guereza*	guereza	5	672	0.00	Grunau & Kuester (2001)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
gumert2000_dom_2	*Macaca fascicularis*	longtailed macaque	7	51	0.00	Gumert (2000)
gunnels1999_dom_1	*Antilocapra americana*	pronghorn	17	223	0.48	Gunnels (1999)
gunnels1999_dom_2	*Antilocapra americana*	pronghorn	12	195	0.23	Gunnels (1999)
hall1983_dom_1	*Cervus elaphus*	red deer	6	153	0.00	Hall (1983)
hanya2008_dom_1	*Macaca fuscata*	Japanese macaque	9	96	0.08	Hanya et al. (2008)
hanya2008_dom_2	*Macaca fuscata*	Japanese macaque	7	22	0.38	Hanya et al. (2008)
harcourt1989_dom_2	*Gorilla gorilla*	gorilla	7	103	0.10	Harcourt & Stewart (1989)
harcourt1989_dom_3	*Gorilla gorilla*	gorilla	6	48	0.07	Harcourt & Stewart (1989)
harcourt1989_dom_4	*Gorilla gorilla*	gorilla	5	37	0.00	Harcourt & Stewart (1989)
harding1986_dom_1	*Erythrocebus patas*	patas monkey	6	152	0.27	Harding & Olson (1986)
hartzler1970_dom_1	*Poecile atricapillus*	black-capped chickadee	7	138	0.10	Hartzler (1970)
hass1991a_dom_1	*Ovis canadensis*	bighorn sheep	20	175	0.50	Hass (1991)
hass1991a_dom_2	*Ovis canadensis*	bighorn sheep	24	459	0.38	Hass (1991)
hass1991a_dom_3	*Ovis canadensis*	bighorn sheep	28	658	0.38	Hass (1991)
hausfater1982_dom_1	*Papio cynocephalus*	yellow baboon	14	1704	0.00	Hausfater et al. (1982)
hausfater1982_dom_2	*Papio cynocephalus*	yellow baboon	20	895	0.04	Hausfater et al. (1982)
hauver2013_dom_2	*Procyon lotor*	raccoon	12	150	0.18	Hauver et al. (2013)
he2013_dom_1	*Rhinopithecus roxellana*	Sichuan snub-nosed monkey	14	1415	0.19	He et al. (2013)
hegner1985_dom_1	*Cyanistes caeruleus*	blue tit	5	77	0.00	Hegner (1985)
hegner1985_dom_2	*Cyanistes caeruleus*	blue tit	5	69	0.00	Hegner (1985)
hegner1985_dom_3	*Cyanistes caeruleus*	blue tit	5	95	0.00	Hegner (1985)
hegner1985_dom_4	*Cyanistes caeruleus*	blue tit	5	82	0.00	Hegner (1985)
heinze1990_dom_1	*Leptothorax*	ant	8	505	0.00	Heinze (1990)
heitor2006_dom_1	*Equus caballus*	Sorraia horse	11	1724	0.00	Heitor & Vicente (2010)
heitor2010_dom_1	*Equus caballus*	Sorraia horse	6	185	0.00	Heitor et al. (2006)
hewitt2009_dom_2	*Meles meles*	European badger	14	122	0.44	Hewitt et al. (2009)
hewitt2009_dom_3	*Meles meles*	European badger	7	164	0.00	Hewitt et al. (2009)
hewitt2009_dom_4	*Meles meles*	European badger	7	118	0.24	Hewitt et al. (2009)
hewitt2009_dom_5	*Meles meles*	European badger	9	198	0.22	Hewitt et al. (2009)
hewitt2009_dom_6	*Meles meles*	European badger	5	37	0.10	Hewitt et al. (2009)
hill1995_dom_1	*Macaca fuscata*	Japanese macaque	7	291	0.00	Hill & Okayasu (1995)
hirotani1994_dom_1	*Rangifer tarandus*	reindeer	13	1557	0.09	Hirotani (1994)
hogstad1987_dom_1	*Poecile montanus*	willow tit	6	130	0.00	Hogstad (1987)
hogstad1987_dom_2	*Poecile montanus*	willow tit	6	43	0.27	Hogstad (1987)
holmes1991_dom_1	*Didelphis virginiana*	Virginia opossum	5	22	0.00	Holmes (1991)
holmes1991_dom_3	*Didelphis virginiana*	Virginia opossum	5	65	0.00	Holmes (1991)
holmes1991_dom_4	*Didelphis virginiana*	Virginia opossum	5	198	0.00	Holmes (1991)
holmes1991_dom_5	*Didelphis virginiana*	Virginia opossum	5	58	0.30	Holmes (1991)
isbell1998_dom_1		vervet monkey	9	127	0.31	Isbell & Pruetz (1998)
isbell1998_dom_2	*Erythrocebus patas*	patas monkey	17	192	0.49	Isbell & Pruetz (1998)
ito1993_dom_1	*Pachycondyla*	ant	8	312	0.00	Ito (1993)
ito1993_dom_2	*Pachycondyla*	ant	8	325	0.04	Ito (1993)
ito1993_dom_3	*Pachycondyla*	ant	15	502	0.12	Ito (1993)
ito1993_dom_4	*Pachycondyla*	ant	15	568	0.13	Ito (1993)
ito1993_dom_6	*Pachycondyla*	ant	15	735	0.03	Ito (1993)
ito1993_dom_7	*Pachycondyla*	ant	8	81	0.14	Ito (1993)
izar2006_dom_1	*Sapajus apella*	brown capuchin	17	325	0.34	Izar et al. (2006)
izawa1980_dom_1	*Sapajus apella*	black-capped capuchin	9	48	0.39	K. Izawa (1980)
izawa2008_dom_1	*Corvus macrorhynchos*	jungle crow	14	441	0.00	E.-I. Izawa & Watanabe (2008)
jacobus1981_dom_1	*Erythrocebus patas*	patas monkey	5	28	0.00	Jacobus & Loy (1981)
jacobus1981_dom_2	*Erythrocebus patas*	patas monkey	5	42	0.00	Jacobus & Loy (1981)
jacobus1981_dom_3	*Erythrocebus patas*	patas monkey	5	43	0.00	Jacobus & Loy (1981)
jaffe2010_dom_1		vervet	8	39	0.46	Jaffe & Isbell (2010)
jaffe2010_dom_3		vervet	7	53	0.19	Jaffe & Isbell (2010)
janson1985_dom_1	*Sapajus apella*	brown capuchin	15	806	0.26	Janson (1985)
jones1980_dom_1	*Alouatta palliata*	mantled howler monkey	18	591	0.14	Jones (1980)
jones1980_dom_2	*Alouatta palliata*	mantled howler monkey	10	242	0.07	Jones (1980)
jones1983_dom_1	*Alouatta caraya*	black howler monkey	5	287	0.00	Jones (1983)
kaburu2012_dom_1	*Macaca sylvanus*	Barbary macaque	7	584	0.00	Kaburu et al. (2012)
katano1985_dom_1	*Nipponocypris temminckii*	dark chub	9	2707	0.11	Katano (1985)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
kaufman1994_dom_01	*Eulemur rufus*	red-fronted lemur	7	44	0.43	Kaufman (1994)
kaufman1994_dom_03	*Eulemur rufus*	red-fronted lemur	7	124	0.10	Kaufman (1994)
kaufman1994_dom_05	*Eulemur rufus*	red-fronted lemur	5	40	0.20	Kaufman (1994)
kaufman1994_dom_07	*Eulemur collaris*	collared brown lemur	7	233	0.10	Kaufman (1994)
kaufman1994_dom_08	*Eulemur collaris*	collared brown lemur	7	71	0.19	Kaufman (1994)
kaufman1994_dom_09	*Eulemur rufus*	red-fronted lemur	10	841	0.00	Kaufman (1994)
kaufman1994_dom_10	*Eulemur rufus*	red-fronted lemur	10	237	0.09	Kaufman (1994)
kaufman1994_dom_11	*Eulemur rufus*	red-fronted lemur	5	232	0.00	Kaufman (1994)
kaufman1994_dom_12	*Eulemur rufus*	red-fronted lemur	5	58	0.10	Kaufman (1994)
keiper1988_dom_1	*Equus przewalskii*	Przewalski horse	9	972	0.25	Keiper (1988)
keiper1992_dom_1	*Equus przewalskii*	Przewalski horse	11	558	0.06	Keiper & Receveur (1992)
keiper1992_dom_2	*Equus przewalskii*	Przewalski horse	7	995	0.00	Keiper & Receveur (1992)
ketterson1979_dom_1	*Junco hyemalis*	dark-eyed junco	17	150	0.47	Ketterson (1979)
kikkawa1980_dom_1	*Zosterops lateralis*	silvereye	10	897	0.07	Kikkawa (1980)
king-bailey2019_dom_1	*Cebus capucinus*	white-faced capuchin	7	49	0.14	King-Bailey (2019)
king-bailey2019_dom_2	*Cebus capucinus*	white-faced capuchin	8	51	0.18	King-Bailey (2019)
king-bailey2019_dom_3	*Cebus capucinus*	white-faced capuchin	8	51	0.21	King-Bailey (2019)
kitchen1974_dom_2	*Antilocapra americana*	pronghorn	6	32	0.40	Kitchen (1974)
kitchen1974_dom_3	*Antilocapra americana*	pronghorn	7	48	0.33	Kitchen (1974)
kitchen1974_dom_6	*Antilocapra americana*	pronghorn	18	531	0.09	Kitchen (1974)
kitchen1974_dom_7	*Antilocapra americana*	pronghorn	15	217	0.15	Kitchen (1974)
knapton1973_dom_1	*Melospiza melodia*	song sparrow	24	1153	0.27	Knapton (1973)
knapton1973_dom_2	*Melospiza melodia*	song sparrow	13	170	0.32	Knapton (1973)
knapton1973_dom_3	*Melospiza melodia*	song sparrow	15	186	0.50	Knapton (1973)
knapton1973_dom_4	*Melospiza melodia*	song sparrow	10	174	0.29	Knapton (1973)
knapton1973_dom_5	*Melospiza melodia*	song sparrow	7	61	0.10	Knapton (1973)
knapton1973_dom_7	*Melospiza melodia*	song sparrow	9	42	0.42	Knapton (1973)
knox1991_dom_1	*Saguinus imperator*	emperor tamarin	9	272	0.11	Knox & Sade (1991)
knox1991_dom_2	*Saguinus imperator*	emperor tamarin	11	291	0.09	Knox & Sade (1991)
koenig2004_dom_2	*Trachypithecus phayrei*	Phayre's leaf monkey	7	68	0.05	Koenig et al. (2004)
kolodziejczyk2005_dom_1	*Haliaeetus albicilla*	white-tailed eagle	13	168	0.44	Kołodziejczyk et al. (2005)
korstjens2002_dom_1	*Colobus polykomos*	black-and-white colobus	9	126	0.11	Korstjens et al. (2002)
koutnik1981_dom_1	*Odocoileus hemionus*	mule deer	8	84	0.11	Koutnik (1981)
koutnik1981_dom_2	*Odocoileus hemionus*	mule deer	8	48	0.18	Koutnik (1981)
kurvers2009_dom_1	*Branta leucopsis*	barnacle goose	18	473	0.22	Kurvers et al. (2009)
lahti1994_dom_1	*Poecile montanus*	willow tit	6	495	0.00	Lahti et al. (1994)
lahti1994_dom_2	*Poecile montanus*	willow tit	5	112	0.00	Lahti et al. (1994)
lahti1994_dom_3	*Poecile montanus*	willow tit	8	335	0.07	Lahti et al. (1994)
langbein2004_dom_1	*Capra hircus*	dwarf goat	12	938	0.00	Langbein & Puppe (2004)
langbein2004_dom_2	*Capra hircus*	dwarf goat	12	782	0.01	Langbein & Puppe (2004)
langbein2004_dom_3	*Sus scrofa domesticus*	pig	10	263	0.04	Langbein & Puppe (2004)
langbein2004_dom_4	*Sus scrofa domesticus*	pig	9	47	0.36	Langbein & Puppe (2004)
lee1979_dom_1	*Papio cynocephalus*	yellow baboon	15	424	0.19	Lee & Oliver (1979)
lee1979_dom_2	*Papio cynocephalus*	yellow baboon	8	173	0.00	Lee & Oliver (1979)
lemasson2006_dom_1	*Cercopithecus campbelli*	Campbell's monkey	8	249	0.21	Lemasson et al. (2006)
lemasson2006_dom_2	*Cercopithecus campbelli*	Campbell's monkey	8	466	0.00	Lemasson et al. (2006)
locati1991_dom_1	*Rupicapra pyrenaica*	Apennine chamois	7	71	0.24	Locati & Lovari (1991)
lott1979_dom_1	*Bison bison*	American bison	26	897	0.32	Lott (1979)
lott1987_dom_1	*Bison bison*	American bison	11	129	0.26	Lott & Galland (1987)
loy1988_dom_1	*Erythrocebus patas*	patas monkey	8	372	0.00	Loy & Harnois (1988)
loy1988_dom_2	*Erythrocebus patas*	patas monkey	8	379	0.04	Loy & Harnois (1988)
loy1988_dom_3	*Erythrocebus patas*	patas monkey	6	185	0.00	Loy & Harnois (1988)
loy1988_dom_4	*Erythrocebus patas*	patas monkey	6	342	0.00	Loy & Harnois (1988)
loy1993_dom_1	*Erythrocebus patas*	patas monkey	6	34	0.27	Loy et al. (1993)
loy1993_dom_2	*Erythrocebus patas*	patas monkey	6	147	0.00	Loy et al. (1993)
lu2008_dom_1	*Semnopithecus entellus*	Hanuman langur	13	137	0.31	Lu et al. (2008)
lu2008_dom_2	*Semnopithecus entellus*	Hanuman langur	13	144	0.32	Lu et al. (2008)
lu2008_dom_3	*Semnopithecus entellus*	Hanuman langur	13	292	0.13	Lu et al. (2008)
lu2008_dom_4	*Semnopithecus entellus*	Hanuman langur	13	379	0.09	Lu et al. (2008)
lu2008_dom_5	*Semnopithecus entellus*	Hanuman langur	13	323	0.10	Lu et al. (2008)
lu2008_dom_6	*Semnopithecus entellus*	Hanuman langur	13	364	0.17	Lu et al. (2008)
marler1955_dom_2	*Fringilla coelebs*	chaffinch	8	197	0.00	Marler (1955)
marsden1968_dom_1	*Macaca mulatta*	rhesus macaque	8	602	0.07	Marsden (1968)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
marsden1968_dom_2	*Macaca mulatta*	rhesus macaque	8	369	0.04	Marsden (1968)
marsden1968_dom_3	*Macaca mulatta*	rhesus macaque	8	328	0.04	Marsden (1968)
marsden1968_dom_4	*Macaca mulatta*	rhesus macaque	6	176	0.07	Marsden (1968)
marsden1968_dom_6	*Macaca mulatta*	rhesus macaque	8	201	0.04	Marsden (1968)
masure1934_dom_1	*Columba livia*	pigeon	7	731	0.00	Masure & Allee (1934b)
masure1934_dom_2	*Columba livia*	pigeon	7	767	0.00	Masure & Allee (1934b)
masure1934a_dom_1	*Melopsittacus undulatus*	shell parrakeet	7	445	0.00	Masure & Allee (1934a)
masure1934a_dom_2	*Melopsittacus undulatus*	shell parrakeet	7	635	0.00	Masure & Allee (1934a)
mather1985_dom_1	*Eledone moschata*	musky octopus	6	98	0.00	Mather (1985)
matsumura1998_dom_1	*Macaca maura*	moor macaque	10	112	0.29	Matsumura (1998)
mccann1981_dom_1	*Mirounga leonina*	southern elephant seal	10	1648	0.02	McCann (1981)
mcmahan1984_dom_1	*Papio anubis*	olive baboon	10	91	0.18	McMahan & Morris (1984)
mehlman1988_dom_1	*Macaca fuscata*	Japanese macaque	15	1202	0.10	Mehlman & Chapais (1988)
meyer1999_dom_1	*Varecia rubra*	red ruffed lemur	5	24	0.40	Meyer et al. (1999)
meyers1993_dom_1	*Propithecus tattersalli*	Tattersall's sifaka	8	269	0.07	Meyers (1993)
meyers1993_dom_2	*Propithecus tattersalli*	Tattersall's sifaka	7	279	0.14	Meyers (1993)
meyers1993_dom_3	*Propithecus tattersalli*	Tattersall's sifaka	7	268	0.00	Meyers (1993)
micheletta2011_dom_1	*Macaca nigra*	crested macaque	7	376	0.05	Micheletta (pers. comm.)
missakian1972_dom_1	*Macaca mulatta*	rhesus macaque	22	5159	0.00	Missakian (1972)
missakian1972_dom_2	*Macaca mulatta*	rhesus macaque	19	349	0.25	Missakian (1972)
missakian1972_dom_3	*Macaca mulatta*	rhesus macaque	16	210	0.28	Missakian (1972)
mitani1993_dom_1	*Pan troglodytes*	chimpanzee	7	128	0.14	Mitani & Nishida (1993)
moller1987_dom_1	*Passer domesticus*	house sparrows	13	429	0.31	Møller (1987)
moller1987_dom_2	*Passer domesticus*	house sparrows	10	690	0.04	Møller (1987)
moller1987_dom_3	*Passer domesticus*	house sparrows	14	578	0.40	Møller (1987)
monnin1999_dom_4	*Dinoponera quadriceps*	ant	6	33	0.33	Monnin & Peeters (1999)
montgomery1957_dom_1	*Equus caballus*	domestic horse	10	488	0.02	Montgomery (1957)
muller-schwarze1979_dom_1	*Rangifer tarandus*	reindeer	8	208	0.04	Müller-Schwarze et al. (1979)
muller-schwarze1979_dom_2	*Rangifer tarandus*	reindeer	8	324	0.25	Müller-Schwarze et al. (1979)
muller2004_dom_1	*Pan troglodytes*	chimpanzee	11	89	0.47	Muller & Wrangham (2004)
muller2004_dom_2	*Pan troglodytes*	chimpanzee	11	127	0.26	Muller & Wrangham (2004)
muma1987_dom_01	*Agelaius phoeniceus*	red-winged blackbird	7	922	0.00	Muma (1987)
muma1987_dom_02	*Agelaius phoeniceus*	red-winged blackbird	6	765	0.00	Muma (1987)
muma1987_dom_03	*Agelaius phoeniceus*	red-winged blackbird	6	528	0.00	Muma (1987)
muma1987_dom_04	*Agelaius phoeniceus*	red-winged blackbird	6	667	0.00	Muma (1987)
muma1987_dom_05	*Agelaius phoeniceus*	red-winged blackbird	9	996	0.06	Muma (1987)
muma1987_dom_06	*Agelaius phoeniceus*	red-winged blackbird	9	1141	0.00	Muma (1987)
muma1987_dom_07	*Agelaius phoeniceus*	red-winged blackbird	8	428	0.00	Muma (1987)
muma1987_dom_08	*Agelaius phoeniceus*	red-winged blackbird	8	859	0.04	Muma (1987)
muma1987_dom_09	*Agelaius phoeniceus*	red-winged blackbird	11	996	0.09	Muma (1987)
muma1987_dom_10	*Agelaius phoeniceus*	red-winged blackbird	11	2054	0.06	Muma (1987)
muma1987_dom_11	*Agelaius phoeniceus*	red-winged blackbird	9	699	0.08	Muma (1987)
muma1987_dom_12	*Agelaius phoeniceus*	red-winged blackbird	9	1877	0.00	Muma (1987)
murray1984_dom_1	*Macaca arctoides*	stumptailed macaque	28	1918	0.11	Murray (1984)
mwamende2009_dom_1	*Cercocebus sanjei*	Sanje mangabey	19	1662	0.14	Mwamende (2009)
myrberg1972_dom_1	*Stegastes partitus*	bicolor damselfish	8	2362	0.00	Myrberg (1972)
myrberg1974_dom_1	*Sphyrna tiburo*	bonnethead shark	10	200	0.02	Myrberg & Gruber (1974)
nakagawa2008_dom_1	*Erythrocebus patas*	patas monkey	6	130	0.00	Nakagawa (2008)
nakagawa2008_dom_2		vervet monkey	8	67	0.36	Nakagawa (2008)
nakano1994_dom_1	*Oncorhynchus masou*	cherry salmon	11	1738	0.04	Nakano (1994)
nakano1995_dom_1	*Salvelinus leucomaenis*	white-spotted charr	14	265	0.22	Nakano (1995)
nakano1995_dom_2	*Oncorhynchus masou*	red-spotted masu salmon	5	99	0.00	Nakano (1995)
nelissen1985_dom_1	*Melanochromis auratus*	auratus cichlid	7	821	0.00	Nelissen (1985)
newton-fisher2004_dom_1	*Pan troglodytes*	chimpanzee	12	134	0.38	Newton-Fisher (2004)
newton-fisher2004_dom_2	*Pan troglodytes*	chimpanzee	12	120	0.33	Newton-Fisher (2004)
norscia2015_dom_01	*Lemur catta*	ringtailed lemur	10	90	0.11	Sclafani et al. (2012); Norscia & Palagi (2015)
norscia2015_dom_02	*Lemur catta*	ringtailed lemur	13	290	0.24	Sclafani et al. (2012); Norscia & Palagi (2015)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
norscia2015_dom_04	*Propithecus verreauxi*	Verreaux's sifaka	6	41	0.20	Palagi et al. (2008); Norscia & Palagi (2015)
norscia2015_dom_06	*Lemur catta*	ringtailed lemur	10	429	0.02	Sclafani et al. (2012); Norscia & Palagi (2015)
norscia2015_dom_07	*Lemur catta*	ringtailed lemur	13	209	0.20	Sclafani et al. (2012); Norscia & Palagi (2015)
norscia2015_dom_08	*Propithecus verreauxi*	Verreaux's sifaka	10	159	0.36	Palagi et al. (2008); Norscia & Palagi (2015)
norscia2015_dom_09	*Propithecus verreauxi*	Verreaux's sifaka	6	69	0.27	Palagi et al. (2008); Norscia & Palagi (2015)
norscia2015_dom_10	*Eulemur fulvus*	brown lemur	11	92	0.31	Norscia & Palagi (2011); Norscia & Palagi (2015)
oi1988_dom_3	*Macaca fuscata*	Japanese macaque	10	85	0.27	Oi (1988)
oi1990_dom_1	*Macaca nemestrina*	pigtailed macaque	22	930	0.18	Oi (1990)
ortius1995_dom_1	*Leptothorax*	ant	6	131	0.00	Ortius & Heinze (1995)
ortius1995_dom_2	*Leptothorax*	ant	5	82	0.30	Ortius & Heinze (1995)
oshea1976_dom_1	*Xerus rutilus*	African unstriped ground squirrel	10	244	0.18	O'Shea (1976)
ostner1999_dom_1		redfronted lemur	6	56	0.27	Ostner & Kappeler (1999)
ostner1999_dom_2		redfronted lemur	9	102	0.47	Ostner & Kappeler (1999)
overdorff2005_dom_1	*Varecia variegata*	black-and-white ruffed lemur	5	75	0.30	Overdorff et al. (2005)
overdorff2005_dom_2	*Varecia variegata*	black-and-white ruffed lemur	5	32	0.30	Overdorff et al. (2005)
owens1996_dom_1	*Parahyaena brunnea*	brown hyaena	7	372	0.24	Owens & Owens (1996)
packer1969_dom_1	*Setonix brachyurus*	quokka	5	275	0.00	Packer (1969)
paoli2008_dom_1	*Pan paniscus*	bonobo	8	300	0.04	Paoli & Palagi (2008)
paoli2008_dom_2	*Pan paniscus*	bonobo	6	246	0.27	Paoli & Palagi (2008)
parsons1980_dom_1	*Zonotrichia leucophrys*	white-crowned sparrow	8	439	0.00	Parsons & Baptista (1980)
parsons1980_dom_2	*Zonotrichia leucophrys*	white-crowned sparrow	8	812	0.00	Parsons & Baptista (1980)
parsons1980_dom_3	*Zonotrichia leucophrys*	white-crowned sparrow	6	178	0.00	Parsons & Baptista (1980)
parsons1980_dom_4	*Zonotrichia leucophrys*	white-crowned sparrow	11	242	0.04	Parsons & Baptista (1980)
paul1989_dom_1	*Macaca sylvanus*	Barbary macaque	5	61	0.10	Paul (1989)
paul1989_dom_2	*Macaca sylvanus*	Barbary macaque	5	12	0.40	Paul (1989)
paul1989_dom_3	*Macaca sylvanus*	Barbary macaque	5	26	0.10	Paul (1989)
paul1989_dom_4	*Macaca sylvanus*	Barbary macaque	5	39	0.10	Paul (1989)
paul1989_dom_5	*Macaca sylvanus*	Barbary macaque	5	9	0.40	Paul (1989)
paul1989_dom_6	*Macaca sylvanus*	Barbary macaque	5	20	0.30	Paul (1989)
paulo2014_dom_1	*Capra hircus*	Saanen goat	17	1472	0.03	Paulo (2014)
paulo2014_dom_2	*Capra hircus*	Saanen goat	14	1194	0.00	Paulo (2014)
payne2003_dom_1	*Cercopithecus albogularis*	samango monkey	9	90	0.08	Payne et al. (2003)
pereira1988_dom_1	*Papio cynocephalus*	yellow baboon	13	1631	0.13	Pereira (1988)
pereira1988_dom_2	*Papio cynocephalus*	yellow baboon	11	1095	0.18	Pereira (1988)
pereira2002_dom_1	*Lemur catta*	ringtailed lemur	11	1463	0.00	Pereira (2002)
perry1996_dom_1	*Cebus capucinus*	white-faced capuchin	6	50	0.07	Perry (1996)
petit1992_dom_1	*Macaca maura*	moor macaque	7	318	0.19	Petit & Thierry (1992)
petit1992_dom_2	*Macaca maura*	moor macaque	7	460	0.00	Petit & Thierry (1992)
pinha2007_dom_1	*Sapajus libidinosus*	bearded capuchin	7	50	0.38	Pinha (2007)
pintodesaalves2013_dom_1	*Inia geoffrensis*	boto	10	824	0.00	Pinto de Sá Alves et al. (2013)
piper1995_dom_1	*Zonotrichia albicollis*	white-throated sparrow	6	1341	0.00	Piper (1995)
piper1995_dom_2	*Zonotrichia albicollis*	white-throated sparrow	7	221	0.00	Piper (1995)
piper1995_dom_3	*Zonotrichia albicollis*	white-throated sparrow	7	391	0.00	Piper (1995)
piper1995_dom_4	*Zonotrichia albicollis*	white-throated sparrow	5	181	0.00	Piper (1995)
poisbleau2005_dom_1	*Anas platyrhynchos*	mallard	8	455	0.04	Poisbleau et al. (2005)
poisbleau2005_dom_2	*Anas acuta*	pintail	6	88	0.13	Poisbleau et al. (2005)
poisbleau2005_dom_3	*Mareca penelope*	wigeon	7	303	0.00	Poisbleau et al. (2005)
poisbleau2006_dom_1	*Anas platyrhynchos*	mallard	16	1160	0.07	Poisbleau et al. (2006)
poisbleau2006_dom_2	*Branta bernicla*	brent goose	19	187	0.49	Poisbleau et al. (2006)
pollock1979_dom_1	*Indri indri*	indri	5	107	0.00	Pollock (1979)
post1992_dom_1	*Quiscalus major*	boat-tailed grackle	15	1163	0.09	Post (1992)
poston1997_dom_1	*Quiscalus major*	boat-tailed grackle	7	200	0.10	Poston (1997)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
prieto1978_dom_1	*Sauromalus ater*	Arizona chuckwalla	18	3253	0.16	Prieto & Ryan (1978)
radesater1974_dom_1	*Branta canadensis*	Canada goose	5	60	0.00	Radesäter (1974)
radesater1974_dom_2	*Branta canadensis*	Canada goose	5	211	0.00	Radesäter (1974)
radesater1974_dom_3	*Branta canadensis*	Canada goose	5	159	0.00	Radesäter (1974)
radesater1974_dom_4	*Anser anser*	greylag goose	5	63	0.00	Radesäter (1974)
radesater1974_dom_5	*Anser anser*	greylag goose	5	202	0.00	Radesäter (1974)
rajpurohit2005_dom_1	*Semnopithecus entellus*	Hanuman langur	6	52	0.13	Rajpurohit (2005)
rajpurohit2005_dom_2	*Semnopithecus entellus*	Hanuman langur	7	75	0.19	Rajpurohit (2005)
range2002_dom_1		sooty mangabey	24	922	0.14	Range & Noë (2002)
raveling1970_dom_1	*Branta canadensis*	Canada goose	5	144	0.00	Raveling (1970)
reason1988_dom_1	*Addax nasomaculatus*	addax	18	425	0.22	Reason & Laird (1988)
reed1997_dom_1	*Macaca nigra*	crested macaque	6	77	0.07	Reed et al. (1997)
reese1982_dom_1	*Pica hudsonia*	black-billed magpie	7	25	0.43	Reese (1982)
rhine1972_dom_1	*Macaca arctoides*	stumptailed macaque	5	166	0.00	Rhine (1972)
rhine1972_dom_2	*Macaca arctoides*	stumptailed macaque	5	221	0.00	Rhine (1972)
rhine1972_dom_3	*Macaca arctoides*	stumptailed macaque	5	245	0.00	Rhine (1972)
rhine1972_dom_4	*Macaca arctoides*	stumptailed macaque	5	379	0.00	Rhine (1972)
rhine1973_dom_1	*Macaca arctoides*	stumptailed macaque	5	122	0.00	Rhine (1973)
rhine1973_dom_2	*Macaca arctoides*	stumptailed macaque	5	255	0.00	Rhine (1973)
rhine1973_dom_3	*Macaca arctoides*	stumptailed macaque	5	74	0.00	Rhine (1973)
rhine1973_dom_4	*Macaca arctoides*	stumptailed macaque	5	221	0.10	Rhine (1973)
richter2007_dom_1	*Macaca arctoides*	stumptailed macaque	9	224	0.06	Richter (2007)
richter2009_dom_1	*Macaca arctoides*	stumptailed macaque	9	258	0.03	Richter et al. (2009)
rizaldi2010_dom_1	*Macaca fuscata*	Japanese macaque	9	412	0.00	Rizaldi & Watanabe (2010)
rizaldi2010_dom_2	*Macaca fuscata*	Japanese macaque	8	200	0.04	Rizaldi & Watanabe (2010)
rizaldi2010_dom_3	*Macaca fuscata*	Japanese macaque	5	123	0.00	Rizaldi & Watanabe (2010)
robbins2008_dom_1	*Gorilla beringei*	gorilla	8	157	0.18	Robbins (2008)
roell1978_dom_1	*Corvus monedula*	jackdaw	18	789	0.16	Röell (1978)
rohwer1985_dom_1	*Zonotrichia querula*	Harris's sparrow	15	530	0.00	Rohwer (1985)
rohwer1985_dom_2	*Zonotrichia querula*	Harris's sparrow	12	304	0.00	Rohwer (1985)
rovero1999_dom_1	*Emys orbicularis*	European pond turtle	8	52	0.32	Rovero et al. (1999)
rovero1999_dom_2	*Emys orbicularis*	European pond turtle	12	75	0.36	Rovero et al. (1999)
russell1970_dom_1	*Osphranter rufus*	red kangaroo	11	143	0.18	Russell (1970)
russell1970_dom_2	*Osphranter rufus*	red kangaroo	13	154	0.31	Russell (1970)
sabine1949_dom_1	*Junco hyemalis*	slate-colored junco	14	149	0.36	Sabine (1949)
sabine1949_dom_2	*Junco hyemalis*	slate-colored junco	11	591	0.07	Sabine (1949)
sabine1949_dom_3	*Spizella arborea*	tree sparrow	6	313	0.07	Sabine (1949)
sabine1959_dom_1	*Junco hyemalis*	Oregon junco	27	2649	0.07	Sabine (1959)
sabine1959_dom_2	*Junco hyemalis*	Oregon junco	18	457	0.28	Sabine (1959)
sabine1959_dom_3	*Junco hyemalis*	Oregon junco	42	4338	0.16	Sabine (1959)
sabine1959_dom_4	*Junco hyemalis*	Oregon junco	26	2169	0.04	Sabine (1959)
sade1972_dom_1	*Macaca mulatta*	rhesus macaque	16	318	0.16	Sade (1972)
samuels1987_dom_2	*Papio cynocephalus*	yellow baboon	19	367	0.23	Samuels et al. (1987)
sapolsky1983_dom_1	*Papio anubis*	olive baboon	14	366	0.13	Sapolsky (1983)
sapolsky1983_dom_2	*Papio anubis*	olive baboon	14	422	0.08	Sapolsky (1983)
satoh2008_dom_2	*Vollenhovia nipponica*	ant	5	102	0.20	Satoh & Ohkawara (2008)
satoh2008_dom_3	*Vollenhovia nipponica*	ant	8	241	0.14	Satoh & Ohkawara (2008)
satoh2008_dom_4	*Vollenhovia nipponica*	ant	9	284	0.08	Satoh & Ohkawara (2008)
savin-williams1977_dom_1	*Homo sapiens*	human	6	214	0.00	Savin-Williams (1977)
savin-williams1980_dom_1	*Homo sapiens*	human	6	879	0.00	Savin-Williams (1980)
savin-williams1980_dom_2	*Homo sapiens*	human	5	705	0.00	Savin-Williams (1980)
scanlon1984_dom_1	*Macaca mulatta*	rhesus macaque	9	116	0.08	Scanlon (1984)
scharf1985_dom_01	*Pica pica*	black-billed magpie	5	37	0.20	Scharf (1985)
scharf1985_dom_02	*Pica pica*	black-billed magpie	5	20	0.10	Scharf (1985)
scharf1985_dom_03	*Pica pica*	black-billed magpie	5	57	0.00	Scharf (1985)
scharf1985_dom_04	*Pica pica*	black-billed magpie	5	30	0.00	Scharf (1985)
scharf1985_dom_05	*Pica pica*	black-billed magpie	5	21	0.20	Scharf (1985)
scharf1985_dom_06	*Pica pica*	black-billed magpie	5	53	0.00	Scharf (1985)
scharf1985_dom_07	*Pica pica*	black-billed magpie	5	18	0.30	Scharf (1985)
scharf1985_dom_08	*Pica pica*	black-billed magpie	5	57	0.00	Scharf (1985)
scharf1985_dom_10	*Pica pica*	black-billed magpie	5	29	0.00	Scharf (1985)
scharf1985_dom_11	*Pica pica*	black-billed magpie	5	75	0.30	Scharf (1985)
scharf1985_dom_12	*Pica pica*	black-billed magpie	5	147	0.00	Scharf (1985)

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data set	species	common name	group size	interactions	sparseness	reference
scharf1985_dom_13	*Pica pica*	black-billed magpie	5	135	0.00	Scharf (1985)
scharf1985_dom_14	*Pica pica*	black-billed magpie	5	51	0.00	Scharf (1985)
scharf1985_dom_15	*Pica pica*	black-billed magpie	5	44	0.10	Scharf (1985)
scharf1985_dom_16	*Pica pica*	black-billed magpie	5	96	0.00	Scharf (1985)
scharf1985_dom_17	*Pica pica*	black-billed magpie	5	60	0.00	Scharf (1985)
scharf1985_dom_18	*Pica pica*	black-billed magpie	5	88	0.20	Scharf (1985)
scharf1985_dom_19	*Pica pica*	black-billed magpie	5	79	0.00	Scharf (1985)
scharf1985_dom_20	*Pica pica*	black-billed magpie	5	119	0.00	Scharf (1985)
scharf1985_dom_21	*Pica pica*	black-billed magpie	5	75	0.00	Scharf (1985)
scharf1985_dom_22	*Pica pica*	black-billed magpie	5	97	0.00	Scharf (1985)
scharf1985_dom_23	*Pica pica*	black-billed magpie	5	99	0.00	Scharf (1985)
scharf1985_dom_24	*Pica pica*	black-billed magpie	5	72	0.10	Scharf (1985)
scharf1985_dom_25	*Pica pica*	black-billed magpie	5	54	0.20	Scharf (1985)
scharf1985_dom_26	*Pica pica*	black-billed magpie	5	37	0.20	Scharf (1985)
scharf1985_dom_27	*Pica pica*	black-billed magpie	5	57	0.10	Scharf (1985)
schein1955_dom_1	*Bos taurus*	dairy cattle	28	495	0.46	Schein & Fohrman (1955)
schneider1984_dom_1	*Zonotrichia albicollis*	white-throated sparrow	10	473	0.00	Schneider (1984)
schusterman1968_dom_1	*Zalophus californianus*	sea lion	7	245	0.33	Schusterman & Dawson (1968)
scott1999_dom_1	*Gorilla gorilla*	gorilla	5	37	0.10	Scott & Lockard (1999)
scott1999_dom_2	*Gorilla gorilla*	gorilla	6	75	0.00	Scott & Lockard (1999)
scott1999_dom_3	*Gorilla gorilla*	gorilla	6	27	0.20	Scott & Lockard (1999)
searcy1979_dom_1	*Agelaius phoeniceus*	red-winged blackbird	6	245	0.00	Searcy (1979)
searcy1979_dom_2	*Agelaius phoeniceus*	red-winged blackbird	5	228	0.00	Searcy (1979)
searcy1979_dom_3	*Agelaius phoeniceus*	red-winged blackbird	5	258	0.00	Searcy (1979)
seibert2001_dom_1	*Nymphicus hollandicus*	cockatiels	12	579	0.04	Seibert & Crowell-Davis (2001)
setchell2001_dom_1	*Mandrillus sphinx*	mandrill	5	66	0.00	Setchell & Dixson (2001)
setchell2001_dom_2	*Mandrillus sphinx*	mandrill	5	58	0.00	Setchell & Dixson (2001)
setchell2001_dom_3	*Mandrillus sphinx*	mandrill	5	47	0.20	Setchell & Dixson (2001)
setchell2001_dom_4	*Mandrillus sphinx*	mandrill	5	17	0.20	Setchell & Dixson (2001)
setchell2005_dom_1	*Mandrillus sphinx*	mandrill	11	1488	0.00	Setchell & Wickings (2005)
setchell2005_dom_2	*Mandrillus sphinx*	mandrill	8	982	0.00	Setchell & Wickings (2005)
seyfarth1976_dom_1	*Papio ursinus*	chacma baboon	8	98	0.11	Seyfarth (1976)
seyfarth1976_dom_2	*Papio ursinus*	chacma baboon	8	140	0.11	Seyfarth (1976)
seyfarth1976_dom_3	*Papio ursinus*	chacma baboon	8	121	0.18	Seyfarth (1976)
seyfarth1980_dom_1		vervet	8	989	0.00	Seyfarth (1980)
seyfarth1980_dom_2		vervet	7	644	0.00	Seyfarth (1980)
seyfarth1980_dom_3		vervet	8	766	0.00	Seyfarth (1980)
shargal2008_dom_1	*Capra nubiana*	ibex	9	469	0.00	Shargal et al. (2008)
sharpe2013_dom_1	*Helogale parvula*	dwarf mongoose	18	297	0.28	Sharpe et al. (2013)
sharpe2013_dom_2	*Helogale parvula*	dwarf mongoose	12	177	0.18	Sharpe et al. (2013)
sharpe2013_dom_3	*Helogale parvula*	dwarf mongoose	18	434	0.24	Sharpe et al. (2013)
sharpe2013_dom_4	*Helogale parvula*	dwarf mongoose	13	280	0.19	Sharpe et al. (2013)
shier2007_dom_1	*Dipodomys heermanni*	Heermann's kangaroo rat	12	350	0.01	Shier & Randall (2007)
shier2007_dom_2	*Dipodomys heermanni*	Heermann's kangaroo rat	12	195	0.00	Shier & Randall (2007)
shoemaker1939_dom_1	*Serinus canaria*	canary	10	10642	0.00	Shoemaker (1939)
silk1981_dom_1	*Macaca radiata*	bonnet macaque	29	1413	0.25	Silk et al. (1981)
singh2003_dom_1	*Macaca fuscata*	Japanese macaque	10	897	0.00	Singh et al. (2003)
singh2011_dom_2	*Macaca radiata*	bonnet macaque	5	121	0.00	Singh et al. (2011)
slotow1993_dom_1	*Zonotrichia leucophrys*	white-crowned sparrow	10	1036	0.02	Slotow et al. (1993)
smith1976_dom_1	*Poecile atricapillus*	black-capped chickadee	7	113	0.05	S. M. Smith (1976)
smith1976_dom_2	*Poecile atricapillus*	black-capped chickadee	7	49	0.14	S. M. Smith (1976)
smith1977_dom_1		squirrel monkey	7	222	0.10	M. Smith et al. (1977)
smith1977_dom_2		squirrel monkey	8	907	0.04	M. Smith et al. (1977)
so2015_dom_1	*Mus*	lab mouse	12	82	0.42	So et al. (2015)
so2015_dom_2	*Mus*	lab mouse	12	166	0.21	So et al. (2015)
so2015_dom_3	*Mus*	lab mouse	12	244	0.18	So et al. (2015)
solberg1997_dom_1	*Passer domesticus*	house sparrow	9	433	0.03	Solberg & Ringsby (1997)
solberg1997_dom_2	*Passer domesticus*	house sparrow	9	271	0.08	Solberg & Ringsby (1997)
solberg1997_dom_3	*Passer domesticus*	house sparrow	9	186	0.22	Solberg & Ringsby (1997)
somers1998_dom_1	*Potamonautes perlatus*	Cape river crab	6	71	0.13	Somers & Nel (1998)
somers1998_dom_2	*Potamonautes perlatus*	Cape river crab	8	128	0.11	Somers & Nel (1998)

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data set	species	common name	group size	interactions	sparseness	reference
somers1998_dom_3	*Potamonautes perlatus*	Cape river crab	9	122	0.11	Somers & Nel (1998)
spratt2019_dom_1	*Alcelaphus buselaphus*	Jackson's hartebeest	10	886	0.07	Spratt et al. (2019)
stahl2000_dom_1	*Macaca silenus*	liontailed macaque	5	606	0.00	Stahl et al. (2000)
stammbach1978_dom_1	*Papio hamadryas*	hamadryas baboon	6	273	0.07	Stammbach (1978)
stammbach1978_dom_2	*Papio hamadryas*	hamadryas baboon	6	572	0.07	Stammbach (1978)
stammbach1982_dom_1	*Papio hamadryas*	hamadryas baboon	6	78	0.07	Stammbach & Kummer (1982)
stammbach1982_dom_2	*Papio hamadryas*	hamadryas baboon	6	93	0.13	Stammbach & Kummer (1982)
stamps1978_dom_1	*Anolis aeneus*	lizard	5	45	0.30	Stamps (1978)
stamps1978_dom_2	*Anolis aeneus*	lizard	6	31	0.27	Stamps (1978)
sterck1997_dom_02	*Macaca fascicularis*	longtailed macaque	9	36	0.42	Sterck & Steenbeek (1997)
struhsaker1967_dom_1		vervet monkey	12	189	0.18	Struhsaker (1967)
struhsaker1967_dom_2		vervet monkey	12	80	0.41	Struhsaker (1967)
strum1982_dom_1	*Papio anubis*	olive baboon	7	1239	0.00	Strum (1982)
strum1982_dom_2	*Papio anubis*	olive baboon	7	110	0.14	Strum (1982)
strum1982_dom_3	*Papio anubis*	olive baboon	7	348	0.05	Strum (1982)
sullivan1981_dom_1	*Phoca vitulina*	harbor seal	5	18	0.20	Sullivan (1981)
sullivan1982_dom_1	*Phoca vitulina*	Pacific harbor seal	18	216	0.37	Sullivan (1982)
surbeck2013_dom_1	*Pan paniscus*	bonobo	21	868	0.24	Surbeck & Hohmann (2013)
tamm1977_dom_1	*Corvus monedula*	jackdaw	5	152	0.00	Tamm (1977)
tamm1977_dom_2	*Corvus monedula*	jackdaw	10	542	0.11	Tamm (1977)
tamura1988_dom_1	*Callosciurus erythraeus*	Formosan squirrel	15	205	0.28	Tamura et al. (1988)
tarvin1997_dom_1	*Cyanocitta cristata*	blue jay	16	553	0.29	Tarvin & Woolfenden (1997)
taub1980_dom_1	*Macaca sylvanus*	Barbary macaque	16	577	0.22	Taub (1980)
taylor1986_dom_1	*Lemur catta*	ringtailed lemur	5	2143	0.00	Taylor (1986)
thompson1960_dom_1	*Haemorhous mexicanus*	house finch	9	989	0.00	W. L. Thompson (1960)
thompson1960_dom_2	*Haemorhous mexicanus*	house finch	8	202	0.00	W. L. Thompson (1960)
thompson1960_dom_3	*Haemorhous mexicanus*	house finch	8	101	0.11	W. L. Thompson (1960)
thompson1960_dom_4	*Haemorhous mexicanus*	house finch	8	103	0.07	W. L. Thompson (1960)
thompson1960_dom_5	*Haemorhous mexicanus*	house finch	8	117	0.00	W. L. Thompson (1960)
thompson1993_dom_1	*Hippotragus niger*	sable antelope	11	794	0.00	K. V. Thompson (1993)
thompson1993_dom_2	*Hippotragus niger*	sable antelope	6	174	0.00	K. V. Thompson (1993)
tilson1984_dom_1	*Crocuta crocuta*	spotted hyaena	8	234	0.32	Tilson & Hamilton (1984)
tilson1988_dom_1	*Equus przewalskii*	Przewalski horse	8	375	0.14	Tilson et al. (1988)
tilson1988_dom_2	*Equus przewalskii*	Przewalski horse	8	302	0.11	Tilson et al. (1988)
tokuda1969_dom_1	*Macaca nemestrina*	pigtailed macaque	16	537	0.12	Tokuda & Jensen (1969)
torr1996_dom_2	*Lampropholis guichenoti*	common garden skink	9	76	0.47	Torr & Shine (1996)
trunzer1999_dom_1	*Neoponera villosa*	ant	19	797	0.09	Trunzer et al. (1999)
tully1975_dom_1	*Macaca radiata*	bonnet macaque	8	247	0.00	Tully (1975)
tully1975_dom_2	*Macaca nemestrina*	pigtailed macaque	8	100	0.11	Tully (1975)
val-laillet2008_dom_1	*Bos taurus*	cow	12	339	0.03	Val-Laillet et al. (2008); supplement to Douglas et al. (2017)
val-laillet2008_dom_2	*Bos taurus*	cow	12	438	0.06	Val-Laillet et al. (2008); supplement to Douglas et al. (2017)
val-laillet2008_dom_3	*Bos taurus*	cow	12	324	0.01	Val-Laillet et al. (2008); supplement to Douglas et al. (2017)
val-laillet2008_dom_4	*Bos taurus*	cow	12	196	0.11	Val-Laillet et al. (2008); supplement to Douglas et al. (2017)
val-laillet2008_dom_5	*Bos taurus*	cow	12	226	0.14	Val-Laillet et al. (2008); supplement to Douglas et al. (2017)
val-laillet2008_dom_6	*Bos taurus*	cow	12	517	0.01	Val-Laillet et al. (2008); supplement to Douglas et al. (2017)
vanderborg2015_dom_1	*Canis familiaris*	domestic dog	10	552	0.09	van der Borg et al. (2015)
vanderborg2015_dom_2	*Canis familiaris*	domestic dog	10	249	0.16	van der Borg et al. (2015)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
vanderborg2015_dom_3	*Canis familiaris*	domestic dog	10	317	0.22	van der Borg et al. (2015)
vandierendonck1995_dom_1	*Equus caballus*	Icelandic horse	21	569	0.03	van Dierendonck et al. (1995)
vanhooff1987_dom_1	*Canis lupus*	wolf	16	10405	0.01	van Hooff & Wensing (1987)
vervaecke2000_dom_1	*Pan paniscus*	bonobo	6	739	0.07	Vervaecke et al. (2000)
vervaecke2007_dom_1	*Equus caballus*	Icelandic horse	9	1039	0.00	Vervaecke et al. (2007)
vervaecke2007_dom_2	*Equus caballus*	Icelandic horse	9	745	0.00	Vervaecke et al. (2007)
vogel2004_dom_1	*Cebus capucinus*	white-faced capuchin	22	306	0.32	Vogel (2004)
vogel2004_dom_2	*Cebus capucinus*	white-faced capuchin	21	217	0.42	Vogel (2004)
vogel2004_dom_3	*Cebus capucinus*	white-faced capuchin	23	532	0.27	Vogel (2004)
watson1970_dom_1	*Passer domesticus*	house sparrow	6	496	0.20	Watson (1970)
watson1970_dom_2	*Passer domesticus*	house sparrow	6	400	0.07	Watson (1970)
watt1982_dom_1	*Zonotrichia leucophrys*	white-crowned sparrow	10	150	0.07	Watt (1982)
watt1982_dom_2	*Zonotrichia leucophrys*	white-crowned sparrow	10	74	0.29	Watt (1982)
watt1985_dom_2	*Spinus tristis*	American goldfinch	6	57	0.27	Watt & Sadowski (1985)
watt1985_dom_4	*Spinus tristis*	American goldfinch	6	58	0.07	Watt & Sadowski (1985)
watt1985_dom_5	*Spinus tristis*	American goldfinch	10	153	0.22	Watt & Sadowski (1985)
watt1986_dom_1	*Zonotrichia querula*	Harris's sparrow	10	511	0.02	Watt (1986)
watt1986_dom_2	*Zonotrichia querula*	Harris's sparrow	10	437	0.02	Watt (1986)
watt1986_dom_3	*Zonotrichia querula*	Harris's sparrow	10	582	0.00	Watt (1986)
watt1986_dom_4	*Zonotrichia querula*	Harris's sparrow	10	576	0.00	Watt (1986)
watt1986_dom_5	*Zonotrichia querula*	Harris's sparrow	10	541	0.02	Watt (1986)
watt1986_dom_6	*Zonotrichia querula*	Harris's sparrow	26	1241	0.15	Watt (1986)
watts1969_dom_1	*Meleagris gallopavo*	turkey	12	508	0.08	C. Watts (1969)
watts1985_dom_1	*Gorilla beringei*	mountain gorilla	5	46	0.00	D. P. Watts (1985)
watts1985_dom_2	*Gorilla beringei*	mountain gorilla	5	19	0.00	D. P. Watts (1985)
watts1985_dom_3	*Gorilla beringei*	mountain gorilla	7	26	0.33	D. P. Watts (1985)
watts1994_dom_1	*Gorilla beringei*	mountain gorilla	11	112	0.07	D. P. Watts (1994)
watts1994_dom_2	*Gorilla beringei*	mountain gorilla	12	132	0.01	D. P. Watts (1994)
watts1994_dom_3	*Gorilla beringei*	mountain gorilla	7	33	0.00	D. P. Watts (1994)
watts1994_dom_4	*Gorilla beringei*	mountain gorilla	5	65	0.00	D. P. Watts (1994)
watts1998_dom_1	*Pan troglodytes*	chimpanzee	24	286	0.46	D. P. Watts (1998)
weeks2000_dom_1	*Equus caballus*	light horse	14	936	0.00	Weeks et al. (2000)
weeks2000_dom_2	*Equus caballus*	light horse	14	139	0.29	Weeks et al. (2000)
weeks2000_dom_3	*Equus caballus*	light horse	14	693	0.02	Weeks et al. (2000)
wells1979_dom_1	*Equus caballus*	Camargue horse	25	3428	0.05	Wells & von Goldschmidt-Rothschild (1979)
wessling2011_dom_1	*Pan troglodytes*	chimpanzee	8	218	0.21	Wessling (2011)
wheatley1978_dom_1	*Macaca fascicularis*	longtailed macaque	6	27	0.13	Wheatley (1978)
whitfield1985_dom_1	*Arenaria interpres*	turnstone	17	276	0.27	Whitfield (1985)
whitten1983_dom_1	*Chlorocebus pygerythrus*	vervet monkey	11	70	0.34	Whitten (1983)
whitten1983_dom_2	*Chlorocebus pygerythrus*	vervet monkey	10	63	0.38	Whitten (1983)
wickings1992_dom_2	*Mandrillus sphinx*	mandrill	6	329	0.27	Wickings & Dixson (1992)
wiedenmann1987_dom_2	*Junco hyemalis*	northern junco	6	204	0.00	Wiedenmann & Rabenold (1987)
wierenga1990_dom_1	*Bos taurus*	dairy cattle	17	3184	0.00	Wierenga (1990)
williams1972_dom_1	*Zosterops lateralis*	greybreasted silvereye	10	1482	0.02	W. T. Williams et al. (1972)
williams1982_dom_3	*Anas platyrhynchos*	mallard	13	297	0.28	D. M. Williams (1982)
williams1982_dom_4	*Anas platyrhynchos*	mallard	13	245	0.30	D. M. Williams (1982)
williams1982_dom_7	*Anas platyrhynchos*	mallard	14	303	0.26	D. M. Williams (1982)
williams1982_dom_8	*Anas platyrhynchos*	mallard	14	425	0.29	D. M. Williams (1982)
wirtz1967_dom_1	*Callospermophilus lateralis*	golden-mantled ground squirrel	8	110	0.29	Wirtz (1967)
wirtz1967_dom_2	*Callospermophilus lateralis*	golden-mantled ground squirrel	10	124	0.49	Wirtz (1967)
wittig2003_dom_1	*Pan troglodytes*	chimpanzee	15	2231	0.16	Wittig & Boesch (2003)
woolfenden1977_dom_1	*Aphelocoma coerulescens*	Florida scrub jay	6	63	0.27	Woolfenden & Fitzpatrick (1977)
woolfenden1977_dom_2	*Aphelocoma coerulescens*	Florida scrub jay	6	25	0.47	Woolfenden & Fitzpatrick (1977)
yamaguchi2008_dom_1	*Columba livia*	pigeon	7	547	0.00	Yamaguchi et al. (2008)

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Table 1: Dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
yamaguchi2008_dom_2	*Columba livia*	pigeon	7	461	0.00	Yamaguchi et al. (2008)
yasukawa1983_dom_01	*Junco hyemalis*	dark-eyed junco	6	547	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_02	*Junco hyemalis*	dark-eyed junco	6	541	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_03	*Junco hyemalis*	dark-eyed junco	6	760	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_04	*Junco hyemalis*	dark-eyed junco	6	853	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_05	*Junco hyemalis*	dark-eyed junco	6	813	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_06	*Junco hyemalis*	dark-eyed junco	6	659	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_07	*Junco hyemalis*	dark-eyed junco	6	752	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_08	*Junco hyemalis*	dark-eyed junco	6	955	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_09	*Junco hyemalis*	dark-eyed junco	6	140	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_10	*Junco hyemalis*	dark-eyed junco	6	218	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_11	*Junco hyemalis*	dark-eyed junco	6	114	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_12	*Junco hyemalis*	dark-eyed junco	6	129	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_13	*Junco hyemalis*	dark-eyed junco	6	265	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_14	*Junco hyemalis*	dark-eyed junco	6	116	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_15	*Junco hyemalis*	dark-eyed junco	6	259	0.00	Yasukawa & Bick (1983)
yasukawa1983_dom_16	*Junco hyemalis*	dark-eyed junco	6	164	0.00	Yasukawa & Bick (1983)
zhang2018_dom_1	*Macaca mulatta*	rhesus macaque	8	195	0.00	Zhang et al. (2018)
zine2000_dom_1	*Ovis canadensis*	mountain sheep	12	381	0.09	Zine & Krausman (2000)
zink1987_dom_1	*Junco hyemalis*	dark-eyed junco	30	1213	0.25	Zink & Watt (1987)

References

Addison, W. E., & Baker, E. (1982). Agonistic behavior and social organization in a herd of goats as affected by the introduction of non-members. *Applied Animal Ethology*, *8*, 527–535. https://doi.org/10.1016/0304-3762(82)90216-4

Alados, C. L., & Escós, J. M. (1992). The determinants of social status and the effect of female rank on reproductive success in Dama and Cuvier's gazelles. *Ethology Ecology & Evolution*, *4*, 151–164. https://doi.org/10.1080/08927014.1992.9525336

Allee, W. C., & Dickinson, J. C. (1954). Dominance and subordination in the smooth dogfish *Mustelus canis* (Mitchill). *Physiological Zoology*, *27*, 356–364. https://doi.org/10.1086/physzool.27.4.30152372

Appleby, M. C. (1983). The probability of linearity in hierarchies. *Animal Behaviour*, *31*, 600–608. https://doi.org/10.1016/S0003-3472(83)80084-0

Archie, E. A., Morrison, T. A., Foley, C. A. H., Moss, C. J., & Alberts, S. C. (2006). Dominance rank relationships among wild female African elephants, *Loxodonta africana*. *Animal Behaviour*, *71*, 117–127. https://doi.org/10.1016/j.anbehav.2005.03.023

Arlet, M. E., & Isbell, L. A. (2009). Variation in behavioral and hormonal responses of adult male gray-cheeked mangabeys (*Lophocebus albigena*) to crowned eagles (*Stephanoaetus coronatus*) in Kibale National Park, Uganda. *Behavioral Ecology and Sociobiology*, *63*, 491. https://doi.org/10.1007/s00265-008-0682-5

Arlet, M. E., Kaasik, A., Molleman, F., Isbell, L. A., Carey, J. R., & Mänd, R. (2011). Social factors increase fecal testosterone levels in wild male gray-cheeked mangabeys (*Lophocebus albigena*). *Hormones and Behavior*, *59*, 605–611. https://doi.org/10.1016/j.yhbeh.2011.02.018

Athman. (2010). Ethology of friendship among adult male bonnet macaques, *Macaca radiata*, at Arunachala Hill, India. In *AMNH*. http://www.amnh.org/learn-teach/young-naturalist-awards/winning-essays2/2010-winning-essays/ethology-of-friendship-among-adult-male-bonnet-macaques-macaca-radiata-at-arunachala-hill-india.

Balasubramaniam, K. N. (2014). *Variation in dominance relationships in macaques from phylogenetic and socioecological perspectives* [PhD thesis]. University at Buffalo, State University of New York.

Banks, E. M. (1956). Social organization in red jungle fowl hens (*Gallus gallus* subsp.). *Ecology*, *37*, 239–248. https://doi.org/10.2307/1933136

Barrette, C., & Vandal, D. (1986). Social rank, dominance, antler size, and access to food in snow-bound wild woodland caribou. *Behaviour*, *97*, 118–145. https://doi.org/10.1163/156853986X00342

Bartlett, D. P., & Meier, G. W. (1971). Dominance status and certain operants in a communal colony of rhesus macaques. *Primates*, *12*, 209–219. https://doi.org/10.1007/BF01730411

Barton, R. A., & Whiten, A. (1993). Feeding competition among female olive baboons, *Papio anubis*. *Animal Behaviour*, *46*, 777–789. https://doi.org/10.1006/anbe.1993.1255

Beilharz, R. G., & Zeeb, K. (1982). Social dominance in dairy cattle. *Applied Animal Ethology*, *8*, 79–97. https://doi.org/10.1016/0304-3762(82)90134-1

Bell, W. J., & Gorton, R. E. (1978). Informational analysis of agonistic behaviour and dominance hierarchy formation in a cockroach, *Nauphoeta cinerea*. *Behaviour*, *67*, 217–235. https://doi.org/10.1163/156853978X00341

Bennett, B. (1986). Social dominance in female bighorn sheep. *Zoo Biology*, *5*, 21–26. https://doi.org/10.1002/zoo.1430050104

Bennett, M. A. (1939). The social hierarchy in ring doves. *Ecology*, *20*, 337–357. https://doi.org/10.2307/1930387

Bercovitch, F. B. (1993). Dominance rank and reproductive maturation in male rhesus macaques (*Macaca mulatta*). *Journal of Reproduction and Fertility*, *99*, 113–120. https://doi.org/10.1530/jrf.0.0990113

Bergerud, A. T. (1974). Rutting behaviour of Newfoundland caribou. In V. Geist & F. R. Walther (Eds.), *The behaviour of ungulates and its relation to management*. International Union for Conservation of Nature; Natural Resources.

Bergstrom, M. L., & Fedigan, L. M. (2010). Dominance among female white-faced capuchin monkeys (*Cebus capucinus*): Hierarchical linearity, nepotism, strength and stability. *Behaviour*, *147*, 899–931. https://doi.org/10.1163/000579510X497283

Berman, C. M., Ionica, C. S., & Li, J. (2004). Dominance style among *Macaca thibetana* on Mt. Huangshan, China. *International Journal of Primatology*, *25*, 1283–1312. https://doi.org/10.1023/B:IJOP.0000043963.77801.c3

Berryman, J. C. (1978). Social behaviour in a colony of domestic guinea pigs: Aggression and dominance. *Zeitschrift Für Tierpsychologie*, *46*, 200–214. https://doi.org/10.1111/j.1439-0310.1978.tb01445.x

Birky, W. A. (2002). *Mating patterns and social structure in a wild group of Formosan macaques (*Macaca cyclopis*)* [PhD thesis]. Rutgers University.

Blatrix, R., & Herbers, J. M. (2004). Intracolonial conflict in the slave-making ant *Protomognathus americanus*: Dominance hierarchies and individual reproductive success. *Insectes Sociaux*, *51*, 131–138. https://doi.org/10.1007/s00040-003-0710-4

Bonanni, R., Cafazzo, S., Fantini, C., Pontier, D., & Natoli, E. (2007). Feeding-order in an urban feral domestic cat colony: Relationship to dominance rank, sex and age. *Animal Behaviour*, *74*, 1369–1379. https://doi.org/10.1016/j.anbehav.2007.02.029

Borries, C., Sommer, V., & Srivastava, A. (1991). Dominance, age, and reproductive success in free-ranging female Hanuman langurs (*Presbytis entellus*). *International Journal of Primatology*, *12*, 231–257. https://doi.org/10.1007/BF02547586

Bromley, P. T. (1991). Manifestations of social dominance in pronghorn bucks. *Applied Animal Behaviour Science*, *29*, 147–164. https://doi.org/10.1016/0168-1591(91)90243-Q

Brown, J. L. (1963). Aggressiveness, dominance and social organization in the Steller jay. *The Condor*, *65*, 460–484. https://doi.org/10.2307/1365507

Bruinzeel, L. W., van de Pol, M., & Trierweiler, C. (2006). Competitive abilities of oystercatchers (*Haematopus ostralegus*) occupying territories of different quality. *Journal of Ornithology*, *147*, 457–463. https://doi.org/10.1007/s10336-005-0037-1

Buck, M. R. (1998). *Female mate choice in sooty mangabeys: Social constraints on mating behavior* [PhD thesis]. Emory University.

Butovskaya, M. L., Kozintsev, A. G., & Welker, C. (1996). Conflict and reconciliation in two groups of crab-eating monkeys differing in social status by birth. *Primates*, *37*, 261–270. https://doi.org/10.1007/BF02381858

Byrne, R. W., Conning, A. M., & Young, J. (1983). Social relationships in a captive group of Diana monkeys (*Cercopithecus diana*). *Primates*, *24*, 360–370. https://doi.org/10.1007/BF02381981

Cafazzo, S., Valsecchi, P., Bonanni, R., & Natoli, E. (2010). Dominance in relation to age, sex, and competitive contexts in a group of free-ranging domestic dogs. *Behavioral Ecology*, *21*, 443–455. https://doi.org/10.1093/beheco/arq001

Canali, E., Verga, M., Montagna, M., & Baldi, A. (1986). Social interactions and induced behavioural reactions in milk-fed female calves. *Applied Animal Behaviour Science*, *16*, 207–215. https://doi.org/10.1016/0168-1591(86)90114-0

Carlisle, T. R., & Zahavi, A. (1986). Helping at the nest, allofeeding and social status in immature Arabian babblers. *Behavioral Ecology and Sociobiology*, *18*, 339–351. https://doi.org/10.1007/BF00299665

Castles, D. L. (1997). *Post-conflict behaviour and relationship quality of cercopithecine primates* [PhD thesis]. University of St. Andrews.

Chacko, S. R. (2014). *Why do females live in groups? Dominance, grooming, and experimental testing of the effects of food distribution on the behavior of commensal bonnet macaque (*Macaca radiata*) females* [PhD thesis]. University of California.

Chancellor, R. L., & Isbell, L. A. (2009). Food site residence time and female competitive relationships in wild gray-cheeked mangabeys (*Lophocebus albigena*). *Behavioral Ecology and Sociobiology*, *63*, 1447–1458. https://doi.org/10.1007/s00265-009-0805-7

Chapais, B. (1985). An experimental analysis of a mother-daughter rank reversal in Japanese macaques (*Macaca fuscata*). *Primates*, *26*, 407–423. https://doi.org/10.1007/BF02382456

Cheney, D. L. (1977). The acquisition of rank and the development of reciprocal alliances among free-ranging immature baboons. *Behavioral Ecology and Sociobiology*, *2*, 303–318. https://doi.org/10.1007/BF00299742

Cheney, D. L., Lee, P. C., & Seyfarth, R. M. (1981). Behavioral correlates of nonrandom mortality among free-ranging female vervet monkeys. *Behavioral Ecology and Sociobiology*, *9*, 153–161. https://doi.org/10.1007/BF00293587

Chism, J. B., & Rogers, W. (1997). Male competition mating success and female choice in a seasonally breeding primate. *Ethology*, *103*, 109–126. https://doi.org/10.1111/j.1439-0310.1997.tb00011.x

Clark, D. L., & Dillon, J. E. (1974). Social dominance relationships between previously unacquainted male and female squirrel monkeys. *Behaviour*, *50*, 218–231. https://doi.org/10.1163/156853974X00462

Clutton-Brock, T. H., Greenwood, P. J., & Powell, R. P. (1976). Ranks and relationships in highland ponies and highland cows. *Zeitschrift Für Tierpsychologie*, *41*, 202–216. https://doi.org/10.1111/j.1439-0310.1976.tb00477.x

Coffin, R. J. (1979). *Attention structure and social organization in three groups of captive vervet monkeys (*Cercopithecus aethiops*)* [PhD thesis]. University of Toronto.

Collias, N. E. (1950). Some variations in grouping and dominance patterns among birds and mammals. *Zoologica*, *35*, 97–119.

Collias, N. E., & Taber, R. D. (1951). A field study of some grouping and dominance relations in ring-necked pheasants. *The Condor*, *53*, 265–275. https://doi.org/10.2307/1364987

Cooper, M. A. (1999). *Social tolerance in Assamese macaques (*Macaca assamensis*)* [PhD thesis]. University of Georgia.

Cords, M. (2000). Agonistic and affiliative relationships in a blue monkey group. In P. F. Whitehead & C. J. Jolly (Eds.), *Old world monkeys* (pp. 453–479). Cambridge University Press. https://doi.org/10.1017/CBO9780511542589.018

Cords, M. (2002). Friendship among adult female blue monkeys (*Cercopithecus mitis*). *Behaviour*, *139*, 291–314. https://doi.org/10.1163/156853902760102681

Côté, S. D. (2000). Dominance hierarchies in female mountain goats: Stability, aggressiveness and determinants of rank. *Behaviour*, *137*, 1541–1566. https://doi.org/10.1163/156853900502718

Coutlee, E. L. (1967). Agonistic behavior in the American goldfinch. *The Wilson Bulletin*, *79*, 89–109.

Cui, L.-W., Sun, Q.-L., & Li, B.-G. (2014). Dominance hierarchy and social relationships in a group of captive black-and-white snub-nosed monkeys (*Rhinopithecus bieti*). *Zoological Research*, *35*, 204. https://doi.org/10.11813/j.issn.0254-5853.2014.3.204

David, B. O., & Stoffels, R. J. (2003). Spatial organisation and behavioural interaction of giant kokopu (*Galaxias argenteus*) in two stream pools differing in fish density. *New Zealand Journal of Marine and Freshwater Research*, *37*, 315–322. https://doi.org/10.1080/00288330.2003.9517169

Dawson, J. W., & Mannan, R. W. (1991). Dominance hierarchies and helper contributions in Harris' hawks. *The Auk*, *108*, 649–660. https://doi.org/10.2307/4088105

de Vries, H. (1995). An improved test of linearity in dominance hierarchies containing unknown or tied relationships. *Animal Behaviour*, *50*, 1375–1389. https://doi.org/10.1016/0003-3472(95)80053-0

de Vries, H., Netto, W. J., & Hanegraaf, P. L. H. (1993). Matman: A program for the analysis of sociometric matrices and behavioural transition matrices. *Behaviour*, *125*, 157–175. https://doi.org/10.1163/156853993X00218

de Vries, H., Stevens, J. M. G., & Vervaecke, H. (2006). Measuring and testing the steepness of dominance hierarchies. *Animal Behaviour*, *71*, 585–592. https://doi.org/10.1016/j.anbehav.2005.05.015

de Waal, F. B. M. (1977). The organization of agonistic relations within two captive groups of Java-monkeys (*Macaca fascicularis*). *Zeitschrift Für Tierpsychologie*, *44*, 225–282. https://doi.org/10.1111/j.1439-0310.1977.tb00995.x

de Waal, F. B. M., & Luttrell, L. M. (1985). The formal hierarchy of rhesus macaques: An investigation of the bared-teeth display. *American Journal of Primatology*, *9*, 73–85. https://doi.org/10.1002/ajp.1350090202

Demaria, C., & Thierry, B. (2001). A comparative study of reconciliation in rhesus and Tonkean macaques. *Behaviour*, *138*, 397–410. https://doi.org/10.1163/15685390152032514

Deng, Z., & Zhao, Q. (1987). Social structure in a wild group of *Macaca thibetana* at Mount Emei, China. *Folia Primatologica*, *49*, 1–10. https://doi.org/10.1159/000156304

Diniz, P., Oliveira, R. S., Marini, M. Â., & Duca, C. (2019). Angry caciques: Intrasexual aggression in a Neotropical colonial blackbird. *Ethology Ecology & Evolution*, *31*, 205–218. https://doi.org/10.1080/03949370.2018.1544593

Dixson, A. F., & Herbert, J. (1977). Testosterone, aggressive behavior and dominance rank in captive adult male talapoin monkeys (*Miopithecus talapoin*). *Physiology & Behavior*, *18*, 534–539. https://doi.org/10.1016/0031-9384(77)90272-4

Douglas, P. H., Ngonga Ngomo, A.-C., & Hohmann, G. (2017). A novel approach for dominance assessment in gregarious species: ADAGIO. *Animal Behaviour*, *123*, 21–32. https://doi.org/10.1016/j.anbehav.2016.10.014

Duboscq, J., Micheletta, J., Agil, M., Hodges, J. K., Thierry, B., & Engelhardt, A. (2013). Social tolerance in wild female crested macaques (*Macaca nigra*) in Tangkoko-Batuangus Nature Reserve, Sulawesi, Indonesia. *American Journal of Primatology*, *75*, 361–375. https://doi.org/10.1002/ajp.22114

Dunford, C. (1970). Behavioral aspects of spatial organization in the chipmunk, *Tamias striatus*. *Behaviour*, *36*, 215–231. https://doi.org/10.1163/156853970X00312

Easley, S. P., Coelho, A. M., Jr, & Taylor, L. L. (1989). Allogrooming, partner choice, and dominance in male anubis baboons. *American Journal of Physical Anthropology*, *80*, 353–368. https://doi.org/10.1002/ajpa.1330800309

Eckert, C. G., & Weatherhead, P. J. (1987). Ideal dominance distributions: A test using red-winged blackbirds (*Agelaius phoeniceus*). *Behavioral Ecology and Sociobiology*, *20*, 43–52. https://doi.org/10.1007/BF00292165

Ellard, M.-E., & Crowell-Davis, S. L. (1989). Evaluating equine dominance in draft mares. *Applied Animal Behaviour Science*, *24*, 55–75. https://doi.org/10.1016/0168-1591(89)90125-1

Ellsworth, E. A., & Belthoff, J. R. (1999). Effects of social status on the dispersal behaviour of juvenile western screech-owls. *Animal Behaviour*, *57*, 883–892. https://doi.org/10.1006/anbe.1998.1050

Erlinge, S. (1977). Agonistic behaviour and dominance in stoats (*Mustela erminea* L.). *Zeitschrift Für Tierpsychologie*, *44*, 375–388. https://doi.org/10.1111/j.1439-0310.1977.tb01002.x

Espmark, Y. (1974). Social behaviour of roe deer at winter feeding stations. *Applied Animal Ethology*, *1*, 35–47. https://doi.org/10.1016/0304-3762(74)90006-6

Fairbanks, W. S. (1994). Dominance, age and aggression among female pronghorn, *Antilocapra americana* (family: Antilocapridae). *Ethology*, *97*, 278–293. https://doi.org/10.1111/j.1439-0310.1994.tb01047.x

Farentinos, R. C. (1972). Social dominance and mating activity in the tassel-eared squirrel (*Sciurus aberti ferreus*). *Animal Behaviour*, *20*, 316–326. https://doi.org/10.1016/S0003-3472(72)80053-8

Feh, C. (1988). Social behaviour and relationships of Prezewalski horses in Dutch semi-reserves. *Applied Animal Behaviour Science*, *21*, 71–87. https://doi.org/10.1016/0168-1591(88)90101-3

Fortunato, M. (2009). *Dominance style in captive, fission-fusioned bonobos (*Pan paniscus*): Dominance hierarchy, aggressive patterns, and conflict management* [PhD thesis]. State University of New York at Buffalo.

Foster, M. S. (1981). Cooperative behavior and social organization of the swallow-tailed manakin (*Chiroxiphia caudata*). *Behavioral Ecology and Sociobiology*, *9*, 167–177. https://doi.org/10.1007/BF00302934

Freeman, N. J. (2012). *Some aspects of male vervet monkey behaviour* [MSc thesis]. University of Lethbridge.

Fugle, G. N., Rothstein, S. I., Osenberg, C. W., & McGinley, M. A. (1984). Signals of status in wintering white-crowned sparrows. *Animal Behaviour*, *32*, 86–93. https://doi.org/10.1016/S0003-3472(84)80327-9

Funkhouser, J. A., Mayhew, J. A., & Mulcahy, J. B. (2018). Social network and dominance hierarchy analyses at Chimpanzee Sanctuary Northwest. *PLOS ONE*, *13*, e0191898. https://doi.org/10.1371/journal.pone.0191898

Furuichi, T. (1997). Agonistic interactions and matrifocal dominance rank of wild bonobos (*Pan paniscus*) at Wamba. *International Journal of Primatology*, *18*, 855–875. https://doi.org/10.1023/A:1026327627943

Gemmill, A. L. (2007). *Adult female feeding competition within two groups of free-ranging ringtailed lemurs (*Lemur catta*) in different habitats at the Beza Mahafaly Special Reserve, southwestern Madagascar* [MSc thesis]. University of Victoria.

Graf, R. P. (1985). Social organization of snowshoe hares. *Canadian Journal of Zoology*, *63*, 468–474. https://doi.org/10.1139/z85-066

Greenberg-Cohen, D., Alkon, P. U., & Yom-Tov, Y. (1994). A linear dominance hierarchy in female Nubian ibex. *Ethology*, *98*, 210–220. https://doi.org/10.1111/j.1439-0310.1994.tb01072.x

Grunau, T., & Kuester, J. (2001). Dominance style in female guerezas (*Colobus guereza* Rüppell 1835). *Primates*, *42*, 301–307. https://doi.org/10.1007/BF02629621

Gumert, M. D. (2000). *Reciprocity in an wild group of long tailed macaques (*Macaca fascicularis*) at Tanjung Puting National Park, Kalimantan Tengah, Indonesia: An analysis of social currency exchange* [MSc thesis]. Bucknell University.

Gunnels, C. W. (1999). *Interaction between winter dominance and territory defense in male pronghorn antelope,* Antilocapra americana [MSc thesis]. Utah State University.

Hall, M. J. (1983). Social organization in an enclosed group of red deer (*Cervus elaphus* L.) On Rhum. I. The dominance hierarchy of females and their offspring. *Zeitschrift Für Tierpsychologie*, *61*, 250–262. https://doi.org/10.1111/j.1439-0310.1983.tb01341.x

Hanya, G., Matsubara, M., Hayaishi, S., Zamma, K., Yoshihiro, S., Kanaoka, M. M., Sugaya, S., Kiyono, M., Nagai, M., Tsuriya, Y., Hayakawa, S., Suzuki, M., Yokota, T., Kondo, D., & Takahata, Y. (2008). Food conditions, competitive regime, and female social relationships in Japanese macaques: Within-population variation on yakushima. *Primates*, *49*, 116–125. https://doi.org/10.1007/s10329-007-0073-y

Harcourt, A. H., & Stewart, K. J. (1989). Functions of alliances in contests within wild gorilla groups. *Behaviour*, *109*, 176–190. https://doi.org/10.1163/156853989X00213

Harding, R. S. O., & Olson, D. K. (1986). Patterns of mating among male patas monkeys (*Erythrocebus patas*) in Kenya. *American Journal of Primatology*, *11*, 343–358. https://doi.org/10.1002/ajp.1350110405

Hartzler, J. E. (1970). Winter dominance relationship in black-capped chickadees. *The Wilson Bulletin*, *82*, 427–434.

Hass, C. C. (1991). Social status in female bighorn sheep (*Ovis canadensis*): Expression, development and reproductive correlates. *Journal of Zoology*, *225*, 509–523. https://doi.org/10.1111/j.1469-7998.1991.tb03832.x

Hausfater, G., Altmann, J., & Altmann, S. A. (1982). Long-term consistency of dominance relations among female baboons (*Papio cynocephalus*). *Science*, *217*, 752–755. https://doi.org/10.1126/science.217.4561.752

Hauver, S., Hirsch, B. T., Prange, S., Dubach, J., & Gehrt, S. D. (2013). Age, but not sex or genetic relatedness, shapes raccoon dominance patterns. *Ethology*, *119*, 769–778. https://doi.org/10.1111/eth.12118

He, H., Zhao, H., Qi, X.-G., Wang, X., Guo, S., Ji, W., Wang, C., Wei, W., & Li, B. (2013). Dominance rank of adult females and mating competition in Sichuan snub-nosed monkeys (*Rhinopithecus roxellana*) in the Qinling Mountains, China. *Chinese Science Bulletin*, *58*, 2205–2211. https://doi.org/10.1007/s11434-012-5649-2

Hegner, R. E. (1985). Dominance and anti-predator behaviour in blue tits (*Parus caeruleus*). *Animal Behaviour*, *33*, 762–768. https://doi.org/10.1016/S0003-3472(85)80008-7

Heinze, J. (1990). Dominance behavior among ant females. *Naturwissenschaften*, *77*, 41–43. https://doi.org/10.1007/BF01131799

Heitor, F., Oom, M. do M., & Vicente, L. M. (2006). Social relationships in a herd of Sorraia horses: Part I. Correlates of social dominance and contexts of aggression. *Behavioural Processes*, *73*, 170–177. https://doi.org/10.1016/j.beproc.2006.05.004

Heitor, F., & Vicente, L. M. (2010). Dominance relationships and patterns of aggression in a bachelor group of Sorraia horses (*Equus caballus*). *Journal of Ethology*, *28*, 35–44. https://doi.org/10.1007/s10164-009-0152-1

Hewitt, S. E., Macdonald, D. W., & Dugdale, H. L. (2009). Context-dependent linear dominance hierarchies in social groups of European badgers, *Meles meles*. *Animal Behaviour*, *77*, 161–169. https://doi.org/10.1016/j.anbehav.2008.09.022

Hill, D. A., & Okayasu, N. (1995). Absence of "youngest ascendancy" in the dominance relations of sisters in wild Japanese macaques (*Macaca fuscata yakui*). *Behaviour*, *132*, 367–379. https://doi.org/10.1163/156853995X00612

Hirotani, A. (1994). Dominance rank, copulatory behaviour and estimated reproductive success in male reindeer. *Animal Behaviour*, *48*, 929–936. https://doi.org/10.1006/anbe.1994.1318

Hogstad, O. (1987). Social rank in winter flocks of willow tits *Parus montanus*. *Ibis*, *129*, 1–9. https://doi.org/10.1111/j.1474-919X.1987.tb03155.x

Holmes, D. J. (1991). Social behavior in captive Virginia opossums, *Didelphis virginiana*. *Journal of Mammalogy*, *72*, 402–410. https://doi.org/10.2307/1382114

Isbell, L. A., & Pruetz, J. D. (1998). Differences between vervets (*Cercopithecus aethiops*) and patas monkeys (*Erythrocebus patas*) in agonistic interactions between adult females. *International Journal of Primatology*, *19*, 837–855. https://doi.org/10.1023/A:1020393329574

Ito, F. (1993). Functional monogyny and dominance hierarchy in the queenless ponerine ant *Pachycondyla* (=*Bothroponera*) sp. In West Java, Indonesia (Hymenoptera, Formicidae, Ponerinae). *Ethology*, *95*, 126–140. https://doi.org/10.1111/j.1439-0310.1993.tb00463.x

Izar, P., Ferreira, R. G., & Sato, T. (2006). Describing the organization of dominance relationships by dominance-directed tree method. *American Journal of Primatology*, *68*, 189–207. https://doi.org/10.1002/ajp.20216

Izawa, E.-I., & Watanabe, S. (2008). Formation of linear dominance relationship in captive jungle crows (*Corvus macrorhynchos*): Implications for individual recognition. *Behavioural Processes*, *78*, 44–52. https://doi.org/10.1016/j.beproc.2007.12.010

Izawa, K. (1980). Social behavior of the wild black-capped capuchin (*Cebus apella*). *Primates*, *21*, 443–467. https://doi.org/10.1007/BF02373834

Jacobus, S., & Loy, J. (1981). The grimace and gecker: A submissive display among patas monkeys. *Primates*, *22*, 393–398. https://doi.org/10.1007/BF02381579

Jaffe, K. E., & Isbell, L. A. (2010). Changes in ranging and agonistic behavior of vervet monkeys (*Cercopithecus aethiops*) after predator-induced group fusion. *American Journal of Primatology*, *72*, 634–644. https://doi.org/10.1002/ajp.20821

Janson, C. H. (1985). Aggressive competition and individual food consumption in wild brown capuchin monkeys (*Cebus apella*). *Behavioral Ecology and Sociobiology*, *18*, 125–138. https://doi.org/10.1007/BF00299041

Jones, C. B. (1980). The functions of status in the mantled howler monkey, *Alouatta palliata* Gray : Intraspecific competition for group membership in a folivorous neotropical primate. *Primates*, *21*, 389–405. https://doi.org/10.1007/BF02390468

Jones, C. B. (1983). Social organization of captive black howler monkeys (*Alouatta caraya*): "Social competition" and the use of non-damaging behavior. *Primates*, *24*, 25–39. https://doi.org/10.1007/BF02381451

Kaburu, S. S. K., MacLarnon, A., Majolo, B., Qarro, M., & Semple, S. (2012). Dominance rank and self-scratching among wild female Barbary macaques (*Macaca sylvanus*). *African Zoology*, *47*, 74–79. https://doi.org/10.3377/004.047.0111

Katano, O. (1985). Aggressive behaviour and dominance relationships of the dark chub, *Zacco temmincki* with special reference to their individual recognition. *Japanese Journal of Ichthyology*, *32*, 225–238. https://doi.org/10.1007/BF02938451

Kaufman, R. (1994). *The nature of agonism and dominance and their relationship to social grooming in the brown lemur,* Eulemur fulvus [PhD thesis, University of Alberta]. https://doi.org/10.7939/R39Z90P91

Keiper, R. R. (1988). Social interactions of the Przewalski horse (*Equus przewalskii* Poliakov, 1881) herd at the Munich Zoo. *Applied Animal Behaviour Science*, *21*, 89–97. https://doi.org/10.1016/0168-1591(88)90102-5

Keiper, R. R., & Receveur, H. (1992). Social interactions of free-ranging Przewalski horses in semi-reserves

in the netherlands. *Applied Animal Behaviour Science*, *33*, 303–318. https://doi.org/10.1016/S0168-1591(05)80068-1

Ketterson, E. D. (1979). Aggressive behavior in wintering dark-eyed juncos: Determinants of dominance and their possible relation to geographic variation in sex ratio. *The Wilson Bulletin*, *91*, 371–383.

Kikkawa, J. (1980). Weight changes in relation to social hierarchy in captive flocks of silvereyes, *Zosterops lateralis*. *Behaviour*, *74*, 92–100. https://doi.org/10.1163/156853980X00320

King-Bailey, G. L. (2019). *The behavior and endocrinology of dominance in female white-faced capuchins (*Cebus capucinus imitator*) in Sector Santa Rosa, Área de Conservación Guanacaste, Costa Rica* [PhD thesis]. Tulane University.

Kitchen, D. W. (1974). Social behavior and ecology of the pronghorn. *Wildlife Monographs*, 3–96.

Knapton, R. W. (1973). *Some ecological aspects of social behaviour in the song sparrow,* Melospiza melodia [MSc thesis, University of British Columbia]. https://doi.org/10.14288/1.0093107

Knox, K. L., & Sade, D. S. (1991). Social behavior of the emperor tamarin in captivity: Components of agonistic display and the agonistic network. *International Journal of Primatology*, *12*, 439–473. https://doi.org/10.1007/BF02547634

Koenig, A., Larney, E., Lu, A., & Borries, C. (2004). Agonistic behavior and dominance relationships in female Phayre's leaf monkeys: Preliminary results. *American Journal of Primatology*, *64*, 351–357. https://doi.org/10.1002/ajp.20084

Kołodziejczyk, M., Kloskowski, J., & Krogulec, J. (2005). Lack of social hierarchy in wintering white-tailed eagles (*Haliaeetus albicilla*) when scavenging. *Ethology Ecology & Evolution*, *17*, 181–188. https://doi.org/10.1080/08927014.2005.9522607

Korstjens, A. H., Sterck, E. H. M., & Noë, R. (2002). How adaptive or phylogenetically inert is primate social behaviour? A test with two sympatric colobines. *Behaviour*, *139*, 203–225. https://doi.org/10.1163/156853902760102654

Koutnik, D. L. (1981). Sex-related differences in the seasonality of agonistic behavior in mule deer. *Journal of Mammalogy*, *62*, 1–11. https://doi.org/10.2307/1380472

Kurvers, R. H. J. M., Eijkelenkamp, B., van Oers, K., van Lith, B., van Wieren, S. E., Ydenberg, R. C., & Prins, H. H. T. (2009). Personality differences explain leadership in barnacle geese. *Animal Behaviour*, *78*, 447–453. https://doi.org/10.1016/j.anbehav.2009.06.002

Lahti, K., Koivula, K., & Orell, M. (1994). Is the social hierarchy always linear in tits? *Journal of Avian Biology*, *25*, 347–348.

Langbein, J., & Puppe, B. (2004). Analysing dominance relationships by sociometric methods - a plea for a more standardised and precise approach in farm animals. *Applied Animal Behaviour Science*, *87*, 293–315. https://doi.org/10.1016/j.applanim.2004.01.007

Lee, P. C., & Oliver, J. I. (1979). Competition, dominance and the acquisition of rank in juvenile yellow baboons (*Papio cynocephalus*). *Animal Behaviour*, *27*, 576–585. https://doi.org/10.1016/0003-3472(79)90193-3

Lemasson, A., Blois-Heulin, C., Jubin, R., & Hausberger, M. (2006). Female social relationships in a captive group of Campbell's monkeys (*Cercopithecus campbelli campbelli*). *American Journal of Primatology*, *68*, 1161–1170. https://doi.org/10.1002/ajp.20315

Locati, M., & Lovari, S. (1991). Clues for dominance in female chamois: Age, weight, or horn size? *Aggressive Behavior*, *17*, 11–15. https://doi.org/10.1002/1098-2337(1991)17:1%3C11::aid-ab2480170103%3E3.0.co;2-#

Lott, D. F. (1979). Dominance relations and breeding rate in mature male American bison. *Zeitschrift Für Tierpsychologie*, *49*, 418–432. https://doi.org/10.1111/j.1439-0310.1979.tb00302.x

Lott, D. F., & Galland, J. C. (1987). Body mass as a factor influencing dominance status in American bison cows. *Journal of Mammalogy*, *68*, 683–685.

Loy, J., Argo, B., Nestell, G.-L., Vallett, S., & Wanamaker, G. (1993). A reanalysis of patas monkeys' "grimace and gecker" display and a discussion of their lack of formal dominance. *International Journal of Primatology*, *14*, 879–893. https://doi.org/10.1007/BF02220257

Loy, J., & Harnois, M. (1988). An assessment of dominance and kinship among patas monkeys. *Primates*, *29*, 331–342. https://doi.org/10.1007/BF02380956

Lu, A., Koenig, A., & Borries, C. (2008). Formal submission, tolerance and socioecological models: A test with female Hanuman langurs. *Animal Behaviour*, *76*, 415–428. https://doi.org/10.1016/j.anbehav.2008.

04.006
Marler, P. (1955). Studies of fighting in chaffinches 1: Behaviour in relation to the social hierarchy. *The British Journal of Animal Behaviour*, *3*, 111–117. https://doi.org/10.1016/S0950-5601(55)80002-0
Marsden, H. M. (1968). Agonistic behaviour of young rhesus monkeys after changes induced in social rank of their mothers. *Animal Behaviour*, *16*, 38–44. https://doi.org/10.1016/0003-3472(68)90106-1
Masure, R. H., & Allee, W. C. (1934a). Flock organization of the shell parrakeet *Melopsittacus undulatus* Shaw. *Ecology*, *15*, 388–398. https://doi.org/10.2307/1932354
Masure, R. H., & Allee, W. C. (1934b). The social order in flocks of the common chicken and the pigeon. *The Auk*, *51*, 306–327. https://doi.org/10.2307/4077659
Mather, J. A. (1985). Behavioural interactions and activity of captive *Eledone moschata*: Laboratory investigations of a "social" octopus. *Animal Behaviour*, *33*, 1138–1144. https://doi.org/10.1016/S0003-3472(85)80173-1
Matsumura, S. (1998). Relaxed dominance relations among female moor macaques (*Macaca maurus*) in their natural habitat, south Sulawesi, Indonesia. *Folia Primatologica*, *69*, 346–356. https://doi.org/10.1159/000021650
McCann, T. S. (1981). Aggression and sexual activity of male southern elephant seals, *Mirounga leonina*. *Journal of Zoology*, *195*, 295–310. https://doi.org/10.1111/j.1469-7998.1981.tb03467.x
McMahan, C. A., & Morris, M. D. (1984). Application of maximum likelihood paired comparison ranking to estimation of a linear dominance hierarchy in animal societies. *Animal Behaviour*, *32*, 374–378. https://doi.org/10.1016/S0003-3472(84)80271-7
Mehlman, P. T., & Chapais, B. (1988). Differential effects of kinship, dominance, and the mating season on female allogrooming in a captive group of *Macaca fuscata*. *Primates*, *29*, 195–217. https://doi.org/10.1007/BF02381122
Meyer, C., Gallo, T., & Schultz, S. T. (1999). Female dominance in captive red ruffed lemurs, *Varecia variegata rubra* (Primates , Lemuridae). *Folia Primatologica*, *70*, 358–361. https://doi.org/10.1159/000021718
Meyers, D. M. (1993). *The effects of resource seasonality on behavior and reproduction in the golden-crowned sifaka (*Propithecus tattersalli*, Simons, 1988) in three Malagasy forests* [PhD thesis]. Duke University.
Missakian, E. A. (1972). Genealogical and cross-genealogical dominance relations in a group of free-ranging rhesus monkeys (*Macaca mulatta*) on Cayo Santiago. *Primates*, *13*, 169–180. https://doi.org/10.1007/BF01840878
Mitani, J. C., & Nishida, T. (1993). Contexts and social correlates of long-distance calling by male chimpanzees. *Animal Behaviour*, *45*, 735–746. https://doi.org/10.1006/anbe.1993.1088
Møller, A. P. (1987). Variation in badge size in male house sparrows *Passer domesticus*: Evidence for status signalling. *Animal Behaviour*, *35*, 1637–1644. https://doi.org/10.1016/S0003-3472(87)80056-8
Monnin, T., & Peeters, C. (1999). Dominance hierarchy and reproductive conflicts among subordinates in a monogynous queenless ant. *Behavioral Ecology*, *10*, 323–332. https://doi.org/10.1093/beheco/10.3.323
Montgomery, G. G. (1957). Some aspects of the sociality of the domestic horse. *Transactions of the Kansas Academy of Science*, *60*, 419–424. https://doi.org/10.2307/3626397
Muller, M. N., & Wrangham, R. W. (2004). Dominance, aggression and testosterone in wild chimpanzees: A test of the "challenge hypothesis." *Animal Behaviour*, *67*, 113–123. https://doi.org/10.1016/j.anbehav.2003.03.013
Müller-Schwarze, D., Källquist, L., & Mossing, T. (1979). Social behavior and chemical communication in reindeer (*Rangifer t. Tarandus* L.). *Journal of Chemical Ecology*, *5*, 483–517. https://doi.org/10.1007/BF00987836
Muma, K. E. (1987). *Sexual selection and plumage variation in female red-winged blackbirds* [MSc thesis]. Carleton University.
Murray, R. D. (1984). *Mating patterns among kin and nonkin in a captive group of stumptail macaques* [PhD thesis]. Ohio State University.
Mwamende, K. A. (2009). *Social organisation, ecology and reproduction in the Sanje mangabey (*Cercocebus sanjei*) in the Udzungwa Mountains National Park, Tanzania* [MSc thesis]. Victoria University.
Myrberg, A. A. (1972). Social dominance and territoriality in the bicolor damselfish, *Eupomacentrus partitus* (Poey) (Pisces: Pomacentridae). *Behaviour*, *41*, 207–230. https://doi.org/10.1163/156853972X00013
Myrberg, A. A., & Gruber, S. H. (1974). The behavior of the bonnethead shark, *Sphyrna tiburo*. *Copeia*,

1974, 358–374. https://doi.org/10.2307/1442530
Nakagawa, N. (2008). Despotic wild patas monkeys (*Erythrocebus patas*) in Kala Maloue, Cameroon. *American Journal of Primatology*, *70*, 238–246. https://doi.org/10.1002/ajp.20481
Nakano, S. (1994). Variation in agonistic encounters in a dominance hierarchy of freely interacting red-spotted masu salmon (*Oncorhynchus masou ishikawai*). *Ecology of Freshwater Fish*, *3*, 153–158. https://doi.org/10.1111/j.1600-0633.1994.tb00017.x
Nakano, S. (1995). Competitive interactions for foraging microhabitats in a size-structured interspecific dominance hierarchy of two sympatric stream salmonids in a natural habitat. *Canadian Journal of Zoology*, *73*, 1845–1854. https://doi.org/10.1139/z95-217
Nelissen, M. H. J. (1985). Structure of the dominance hierarchy and dominance determining "group factors" in *Melanochromis auratus* (Pisces, Cichlidae). *Behaviour*, *94*, 85–107. https://doi.org/10.1163/156853985X00280
Newton-Fisher, N. E. (2004). Hierarchy and social status in Budongo chimpanzees. *Primates*, *45*, 81–87. https://doi.org/10.1007/s10329-003-0064-6
Norscia, I., & Palagi, E. (2011). Do wild brown lemurs reconcile? Not always. *Journal of Ethology*, *29*, 181–185. https://doi.org/10.1007/s10164-010-0228-y
Norscia, I., & Palagi, E. (2015). The socio-matrix reloaded: From hierarchy to dominance profile in wild lemurs. *PeerJ*, *3*, e729. https://doi.org/10.7717/peerj.729
O'Shea, T. J. (1976). Home range, social behavior, and dominance relationships in the African unstriped ground squirrel, *Xerus rutilus*. *Journal of Mammalogy*, *57*, 450–460. https://doi.org/10.2307/1379295
Oi, T. (1988). Sociological study on the troop fission of wild Japanese monkeys (*Macaca fuscata yakui*) on Yakushima Island. *Primates*, *29*, 1–19. https://doi.org/10.1007/BF02380846
Oi, T. (1990). Patterns of dominance and affiliation in wild pig-tailed macaques (*Macaca nemestrina nemestrina*) in west Sumatra. *International Journal of Primatology*, *11*, 339–356. https://doi.org/10.1007/BF02193005
Ortius, D., & Heinze, J. (1995). Dynamics and consequences of hierarchy formation in the ant *Leptothorax* sp. A. *Ethology*, *99*, 223–233. https://doi.org/10.1111/j.1439-0310.1995.tb00896.x
Ostner, J., & Kappeler, P. M. (1999). Central males instead of multiple pairs in redfronted lemurs, *Eulemur fulvus rufus* (Primates, Lemuridae)? *Animal Behaviour*, *58*, 1069–1078. https://doi.org/10.1006/anbe.1999.1222
Overdorff, D. J., Erhart, E. M., & Mutschler, T. (2005). Does female dominance facilitate feeding priority in black-and-white ruffed lemurs (*Varecia variegata*) in southeastern Madagascar? *American Journal of Primatology*, *66*, 7–22. https://doi.org/10.1002/ajp.20125
Owens, D., & Owens, M. (1996). Social dominance and reproductive patterns in brown hyaenas, *Hyaena brunnea*, of the central Kalahari desert. *Animal Behaviour*, *51*, 535–551. https://doi.org/10.1006/anbe.1996.0058
Packer, W. C. (1969). Observations on the behavior of the marsupial *Setonix brachyurus* (Quoy and Gaimard) in an enclosure. *Journal of Mammalogy*, *50*, 8–20. https://doi.org/10.2307/1378625
Palagi, E., Antonacci, D., & Norscia, I. (2008). Peacemaking on treetops: First evidence of reconciliation from a wild prosimian (*Propithecus verreauxi*). *Animal Behaviour*, *76*, 737–747. https://doi.org/10.1016/j.anbehav.2008.04.016
Paoli, T., & Palagi, E. (2008). What does agonistic dominance imply in bonobos? In T. Furuichi & J. Thompson (Eds.), *The bonobos* (pp. 39–54). Springer. https://doi.org/10.1007/978-0-387-74787-3_3
Parsons, J., & Baptista, L. F. (1980). Crown color and dominance in the white-crowned sparrow. *The Auk*, *97*, 807–815. https://doi.org/10.1093/auk/97.4.807
Paul, A. (1989). Determinants of male mating success in a large group of Barbary macaques (*Macaca sylvanus*) at Affenberg Salem. *Primates*, *30*, 461–476. https://doi.org/10.1007/BF02380874
Paulo, J. L. de A. (2014). *Caracterização comportamental e avaliação da qualidade de leite em caprinos da raça Saanen no Semiárido nordestino* [Tese (Doutorado)]. Universidade Federal do Rio Grande do Norte.
Payne, H. F. P., Lawes, M. J., & Henzi, S. P. (2003). Competition and the exchange of grooming among female samango monkeys (*Cercopithecus mitis erythrachus*). *Behaviour*, *140*, 453–471. https://doi.org/10.1163/156853903322127931
Pereira, M. E. (1988). Agonistic interactions of juvenile savanna baboons. *Ethology*, *79*, 195–217. https://doi.org/10.1111/j.1439-0310.1988.tb00711.x

Pereira, M. E. (2002). Agonistic interaction, dominance relation, and ontogenetic trajectories in ringtailed lemurs. In M. E. Pereira & L. A. Fairbanks (Eds.), *Juvenile primates: Life history, development, and behavior* (pp. 285–305). University of Chicago Press.

Perry, S. E. (1996). Female-female social relationships in wild white-faced capuchin monkeys, *Cebus capuchinus*. *American Journal of Primatology*, *40*, 167–182. https://doi.org/10.1002/(SICI)1098-2345(1996)40:2%3C167::AID-AJP4%3E3.0.CO;2-W

Petit, O., & Thierry, B. (1992). Affiliative function of the silent bared-teeth display in moor macaques (*Macaca maurus*): Further evidence for the particular status of Sulawesi macaques. *International Journal of Primatology*, *13*, 97–105. https://doi.org/10.1007/BF02547729

Pinha, P. S. (2007). *Interações sociais em grupos de macacos-prego (*Cebus libidinosus*) no Parque Nacional de Brasília* [MSc thesis]. Universidade de Brasília.

Pinto de Sá Alves, L. C., Andriolo, A., Orams, M. B., & de Freitas Azevedo, A. (2013). Resource defence and dominance hierarchy in the boto (*Inia geoffrensis*) during a provisioning program. *Acta Ethologica*, *16*, 9–19. https://doi.org/10.1007/s10211-012-0132-2

Piper, W. H. (1995). Social dominance in young white-throated sparrows: Effects of early social experience and the unstable period. *The Auk*, *112*, 878–889. https://doi.org/10.2307/4089020

Poisbleau, M., Fritz, H., Guillemain, M., & Lacroix, A. (2005). Testosterone and linear social dominance status in captive male dabbling ducks in winter. *Ethology*, *111*, 493–509. https://doi.org/10.1111/j.1439-0310.2005.01092.x

Poisbleau, M., Jenouvrier, S., & Fritz, H. (2006). Assessing the reliability of dominance scores for assigning individual ranks in a hierarchy. *Animal Behaviour*, *72*, 835–842. https://doi.org/10.1016/j.anbehav.2006.01.024

Pollock, J. I. (1979). Female dominance in *Indri indri*. *Folia Primatologica*, *31*, 143–164. https://doi.org/10.1159/000155877

Post, W. (1992). Dominance and mating success in male boat-tailed grackles. *Animal Behaviour*, *44*, 917–929. https://doi.org/10.1016/S0003-3472(05)80587-1

Poston, J. P. (1997). Dominance, access to colonies, and queues for mating opportunities by male boat-tailed grackles. *Behavioral Ecology and Sociobiology*, *41*, 89–98. https://doi.org/10.1007/s002650050368

Prieto, A. A., & Ryan, M. J. (1978). Some observations of the social behavior of the Arizona chuckwalla, *Sauromalus obesus tumidus* (Reptilia, Lacertilia, Iguanidae). *Journal of Herpetology*, *12*, 327–336. https://doi.org/10.2307/1563613

Radesäter, T. (1974). On the ontogeny of orienting movements in the triumph ceremony in two species of geese (*Anser anser* L. And *Branta canadensis* L.). *Behaviour*, *50*, 1–15. https://doi.org/10.1163/156853974X00011

Rajpurohit, D. S. (2005). *Study the dominance hierarchy and its role in social organization in Hanuman langur,* Semnopithecus entellus entellus [PhD thesis]. Jai Narain Vyas University.

Range, F., & Noë, R. (2002). Familiarity and dominance relations among female sooty mangabeys in the Taï National Park. *American Journal of Primatology*, *56*, 137–153. https://doi.org/10.1002/ajp.1070

Raveling, D. G. (1970). Dominance relationships and agonistic behavior of Canada geese in winter. *Behaviour*, *37*, 291–318. https://doi.org/10.1163/156853970X00394

Reason, R. C., & Laird, E. W. (1988). Determinants of dominance in captive female addax (*Addax nasomaculatus*). *Journal of Mammalogy*, *69*, 375–377. https://doi.org/10.2307/1381391

Reed, C., O'Brien, T. G., & Kinnaird, M. F. (1997). Male social behavior and dominance hierarchy in the Sulawesi crested black macaque (*Macaca nigra*). *International Journal of Primatology*, *18*, 247–260. https://doi.org/10.1023/A:1026376720249

Reese, K. P. (1982). *The influence of winter social behavior on the habitat selection and reproductive success of the black-billed magpie* [PhD thesis]. Utah State University.

Rhine, R. J. (1972). Changes in the social structure of two groups of stumptail macaques (*Macaca arctoides*). *Primates*, *13*, 181–194. https://doi.org/10.1007/BF01840879

Rhine, R. J. (1973). Variation and consistency in the social behavior of two groups of stumptail macaques (*Macaca arctoides*). *Primates*, *14*, 21–35. https://doi.org/10.1007/BF01730513

Richter, C. (2007). *Social relationships in free-ranging male stumptail macaques (*Macaca arctoides*)* [Diploma Thesis]. University Leipzig.

Richter, C., Mevis, L., Malaivijitnond, S., Schülke, O., & Ostner, J. (2009). Social relationships in free-

ranging male *Macaca arctoides*. *International Journal of Primatology*, *30*, 625–642. https://doi.org/10.1007/s10764-009-9364-z
Rizaldi, & Watanabe, K. (2010). Early development of peer dominance relationships in a captive group of Japanese macaques *Macaca fuscata*. *Current Zoology*, *56*, 190–197.
Robbins, M. M. (2008). Feeding competition and agonistic relationships among Bwindi *Gorilla beringei*. *International Journal of Primatology*, *29*, 999. https://doi.org/10.1007/s10764-008-9275-4
Röell, A. (1978). Social behaviour of the jackdaw, *Corvus monedula*, in relation to its niche. *Behaviour*, *64*, 1–124. https://doi.org/10.1163/156853978X00459
Rohwer, S. (1985). Dyed birds achieve higher social status than controls in Harris' sparrows. *Animal Behaviour*, *33*, 1325–1331. https://doi.org/10.1016/S0003-3472(85)80193-7
Rovero, F., Lebboroni, M., & Chelazzi, G. (1999). Aggressive interactions and mating in wild populations of the European pond turtle *Emys orbicularis*. *Journal of Herpetology*, *33*, 258–263. https://doi.org/10.2307/1565723
Russell, E. M. (1970). Agonistic interactions in the red kangaroo (*Megaleia rufa*). *Journal of Mammalogy*, *51*, 80–88. https://doi.org/10.2307/1378534
Sabine, W. S. (1949). Dominance in winter flocks of juncos and tree sparrows. *Physiological Zoology*, *22*, 64–85.
Sabine, W. S. (1959). The winter society of the Oregon junco: Intolerance, dominance, and the pecking order. *The Condor*, *61*, 110–135. https://doi.org/10.2307/1365213
Sade, D. S. (1972). Sociometrics of *Macaca mulatta* I. Linkages and cliques in grooming matrices. *Folia Primatologica*, *18*, 196–223. https://doi.org/10.1159/000155480
Samuels, A., Silk, J. B., & Altmann, J. (1987). Continuity and change in dominance relations among female baboons. *Animal Behaviour*, *35*, 785–793. https://doi.org/10.1016/S0003-3472(87)80115-X
Sapolsky, R. M. (1983). Endocrine aspects of social instability in the olive baboon (*Papio anubis*). *American Journal of Primatology*, *5*, 365–379. https://doi.org/10.1002/ajp.1350050406
Satoh, A., & Ohkawara, K. (2008). Dominance hierarchies and aggressive behavior among queens of the inquiline ant *Vollenhovia nipponica*. *Insectes Sociaux*, *55*, 200–206. https://doi.org/10.1007/s00040-008-0989-2
Savin-Williams, R. C. (1977). Dominance in a human adolescent group. *Animal Behaviour*, *25*, 400–406. https://doi.org/10.1016/0003-3472(77)90014-8
Savin-Williams, R. C. (1980). Dominance hierarchies in groups of middle to late adolescent males. *Journal of Youth and Adoloscence*, *9*, 75–85. https://doi.org/10.1007/BF02088381
Scanlon, K. (1984). *Play and development of social skills in infant rhesus macaques* [PhD thesis, Open University]. https://doi.org/10.21954/ou.ro.0000de55
Scharf, C. S. (1985). *The role of social dominance in winter flocks of black-billed magpies* [PhD thesis, University of Alberta]. https://doi.org/10.7939/R3707X058
Schein, M. W., & Fohrman, M. H. (1955). Social dominance relationships in a herd of dairy cattle. *The British Journal of Animal Behaviour*, *3*, 45–55. https://doi.org/10.1016/S0950-5601(55)80012-3
Schneider, K. J. (1984). Dominance, predation, and optimal foraging in white-throated sparrow flocks. *Ecology*, *65*, 1820–1827. https://doi.org/10.2307/1937778
Schusterman, R. J., & Dawson, R. G. (1968). Barking, dominance, and territoriality in male sea lions. *Science*, *160*, 434–436. https://doi.org/10.1126/science.160.3826.434
Sclafani, V., Norscia, I., Antonacci, D., & Palagi, E. (2012). Scratching around mating: Factors affecting anxiety in wild *Lemur catta*. *Primates*, *53*, 247–254. https://doi.org/10.1007/s10329-012-0294-6
Scott, J. A., & Lockard, J. S. (1999). Female dominance relationships among captive western lowland gorillas: Comparisons with the wild. *Behaviour*, *136*, 1283–1310. https://doi.org/10.1163/156853999500721
Searcy, W. A. (1979). Morphological correlates of dominance in captive male red-winged blackbirds. *The Condor*, *81*, 417–420. https://doi.org/10.2307/1366971
Seibert, L. M., & Crowell-Davis, S. L. (2001). Gender effects on aggression, dominance rank, and affiliative behaviors in a flock of captive adult cockatiels (*Nymphicus hollandicus*). *Applied Animal Behaviour Science*, *71*, 155–170. https://doi.org/10.1016/S0168-1591(00)00172-6
Setchell, J. M., & Dixson, A. F. (2001). Arrested development of secondary sexual adornments in subordinate adult male mandrills (*Mandrillus sphinx*). *American Journal of Physical Anthropology*, *115*, 245–252. https://doi.org/10.1002/ajpa.1079

Setchell, J. M., & Wickings, E. J. (2005). Dominance, status signals and coloration in male mandrills (*Mandrillus sphinx*). *Ethology*, *111*, 25–50. https://doi.org/10.1111/j.1439-0310.2004.01054.x
Seyfarth, R. M. (1976). Social relationships among adult female baboons. *Animal Behaviour*, *24*, 917–938. https://doi.org/10.1016/S0003-3472(76)80022-X
Seyfarth, R. M. (1980). The distribution of grooming and related behaviours among adult female vervet monkeys. *Animal Behaviour*, *28*, 798–813. https://doi.org/10.1016/S0003-3472(80)80140-0
Shargal, D., Shore, L., Roteri, N., Terkel, A., Zorovsky, Y., Shemesh, M., & Steinberger, Y. (2008). Fecal testosterone is elevated in high ranking female ibexes (*Capra nubiana*) and associated with increased aggression and a preponderance of male offspring. *Theriogenology*, *69*, 673–680. https://doi.org/10.1016/j.theriogenology.2007.11.017
Sharpe, L. L., Hill, A., & Cherry, M. I. (2013). Individual recognition in a wild cooperative mammal using contact calls. *Animal Behaviour*, *86*, 893–900. https://doi.org/10.1016/j.anbehav.2013.07.023
Shier, D. M., & Randall, J. A. (2007). Use of different signaling modalities to communicate status by dominant and subordinate heermann's kangaroo rats (*Dipodomys heermanni*). *Behavioral Ecology and Sociobiology*, *61*, 1023–1032. https://doi.org/10.1007/s00265-006-0335-5
Shizuka, D., & McDonald, D. B. (2012). A social network perspective on measurements of dominance hierarchies. *Animal Behaviour*, *83*, 925–934. https://doi.org/10.1016/j.anbehav.2012.01.011
Shoemaker, H. H. (1939). Social hierarchy in flocks of the canary. *The Auk*, *56*, 381–406. https://doi.org/10.2307/4078790
Silk, J. B., Samuels, A., & Rodman, P. S. (1981). Hierarchical organization of female *Macaca radiata* in captivity. *Primates*, *22*, 84–95. https://doi.org/10.1007/BF02382559
Singh, M., Jeyaraj, T., Prashanth, U., & Kaumanns, W. (2011). Male–male relationships in lion-tailed macaques (*Macaca silenus*) and bonnet macaques (*Macaca radiata*). *International Journal of Primatology*, *32*, 167–176. https://doi.org/10.1007/s10764-010-9448-9
Singh, M., Singh, M., Sharma, A. K., & Krishna, B. A. (2003). Methodological considerations in measurement of dominance in primates. *Current Science*, *84*, 709–713.
Slotow, R., Alcock, J., & Rothstein, S. I. (1993). Social status signalling in white-crowned sparrows: An experimental test of the social control hypothesis. *Animal Behaviour*, *46*, 977–989. https://doi.org/10.1006/anbe.1993.1279
Smith, M., Harris, P. J., & Strayer, F. F. (1977). Laboratory methods for the assessment of social dominance among captive squirrel monkeys. *Primates*, *18*, 977–984. https://doi.org/10.1007/BF02382948
Smith, S. M. (1976). Ecological aspects of dominance hierarchies in black-capped chickadees. *The Auk*, *93*, 95–107. https://doi.org/10.1093/auk/93.1.95
So, N., Franks, B., Lim, S., & Curley, J. P. (2015). A social network approach reveals associations between mouse social dominance and brain gene expression. *PLoS ONE*, *10*, e0134509. https://doi.org/10.1371/journal.pone.0134509
Solberg, E. J., & Ringsby, T. H. (1997). Does male badge size signal status in small island populations of house sparrows, *Passer domesticus*? *Ethology*, *103*, 177–186. https://doi.org/10.1111/j.1439-0310.1997.tb00114.x
Somers, M. J., & Nel, J. A. J. (1998). Dominance and population structure of freshwater crabs (*Potamonautes perlatus* Milne Edwards). *South African Journal of Zoology*, *33*, 31–36. https://doi.org/10.1080/02541858.1998.11448450
Spratt, K. S., Spratt, J. S., Bauman, J. E., & Chandler, C. R. (2019). Behavioral and endocrine correlates of dominance in captive female Jackson's hartebeest (*Alcelaphus buselaphus*). *Zoo Biology*, *38*, 157–166. https://doi.org/10.1002/zoo.21462
Stahl, D., Herrmann, F., & Kaumanns, W. (2000). Group formation of a captive all male group of lion-tailed macaques (*Macaca silenus*). *Primate Report*, *58*, 93–108.
Stammbach, E. (1978). On social differentiation in groups of captive female hamadryas baboons. *Behaviour*, *67*, 322–338. https://doi.org/10.1163/156853978X00387
Stammbach, E., & Kummer, H. (1982). Individual contributions to a dyadic interaction: An analysis of grooming. *Animal Behaviour*, *30*, 964–971. https://doi.org/10.1016/S0003-3472(82)80184-X
Stamps, J. A. (1978). A field study of the ontogeny of social behavior in the lizard *Anolis aeneus*. *Behaviour*, *66*, 1–30. https://doi.org/10.1163/156853978X00396
Sterck, E. H. M., & Steenbeek, R. (1997). Female dominance relationships and food competition in the

sympatric Thomas langur and long-tailed macaque. *Behaviour*, *134*, 749–774. https://doi.org/10.1163/156853997X00052

Struhsaker, T. T. (1967). Social structure among vervet monkeys (*Cercopithecus aethiops*). *Behaviour*, *29*, 83–121. https://doi.org/10.1163/156853967X00073

Strum, S. C. (1982). Agonistic dominance in male baboons: An alternative view. *International Journal of Primatology*, *3*, 175–202. https://doi.org/10.1007/BF02693494

Sullivan, R. M. (1981). Aquatic displays and interactions in harbor seals, *Phoca vitulina*, with comments on mating systems. *Journal of Mammalogy*, *62*, 825–831. https://doi.org/10.2307/1380606

Sullivan, R. M. (1982). Agonistic behavior and dominance relationships in the harbor seal, *Phoca vitulina*. *Journal of Mammalogy*, *63*, 554–569. https://doi.org/10.2307/1380260

Surbeck, M., & Hohmann, G. (2013). Intersexual dominance relationships and the influence of leverage on the outcome of conflicts in wild bonobos (*Pan paniscus*). *Behavioral Ecology and Sociobiology*, *67*, 1767–1780. https://doi.org/10.1007/s00265-013-1584-8

Tamm, S. (1977). Social dominance in captive jackdaws (*Corvus monedula*). *Behavioural Processes*, *2*, 293–299. https://doi.org/10.1016/0376-6357(77)90032-8

Tamura, N., Hayashi, F., & Miyashita, K. (1988). Dominance hierarchy and mating behavior of the Formosan squirrel, *Callosciurus erythraeus thaiwanensis*. *Journal of Mammalogy*, *69*, 320–331. https://doi.org/10.2307/1381382

Tarvin, K. A., & Woolfenden, G. E. (1997). Patterns of dominance and aggressive behavior in blue jays at a feeder. *Condor*, *99*, 434–444.

Taub, D. M. (1980). Testing the "agonistic buffering" hypothesis. *Behavioral Ecology and Sociobiology*, *6*, 187–197. https://doi.org/10.1007/BF00569200

Taylor, L. L. (1986). *Kinship, dominance, and social organization in a semi-free ranging group of ringtailed lemurs (*Lemur catta*)* [PhD thesis]. Washington University.

Thompson, K. V. (1993). Aggressive behavior and dominance hierarchies in female sable antelope, *Hippotragus niger*: Implications for captive management. *Zoo Biology*, *12*, 189–202. https://doi.org/10.1002/zoo.1430120205

Thompson, W. L. (1960). Agonistic behavior in the house finch - part II: Factors in aggressiveness and sociality. *The Condor*, *62*, 378–402. https://doi.org/10.2307/1365167

Tilson, R. L., & Hamilton, W. J., III. (1984). Social dominance and feeding patterns of spotted hyaenas. *Animal Behaviour*, *32*, 715–724. https://doi.org/10.1016/S0003-3472(84)80147-5

Tilson, R. L., Sweeny, K. A., Binczik, G. A., & Reindl, N. J. (1988). Buddies and bullies: Social structure of a bachelor group of Przewalski horses. *Applied Animal Behaviour Science*, *21*, 169–185. https://doi.org/10.1016/0168-1591(88)90106-2

Tokuda, K., & Jensen, G. D. (1969). Determinants of dominance hierarchy in a captive group of pigtailed monkeys (*Macaca nemestrina*). *Primates*, *10*, 227–236. https://doi.org/10.1007/BF01730344

Torr, G. A., & Shine, R. (1996). Patterns of dominance in the small scincid lizard *Lampropholis guichenoti*. *Journal of Herpetology*, *30*, 230. https://doi.org/10.2307/1565514

Trunzer, B., Heinze, J., & Hölldobler, B. (1999). Social status and reproductive success in queenless ant colonies. *Behaviour*, *136*, 1093–1105. https://doi.org/10.1163/156853999501775

Tully, M. S. (1975). *Social relations among adult females in two species of macaque* [PhD thesis]. University of Colorado.

Val-Laillet, D., de Passillé, A. M., Rushen, J., & von Keyserlingk, M. A. G. (2008). The concept of social dominance and the social distribution of feeding-related displacements between cows. *Applied Animal Behaviour Science*, *111*, 158–172. https://doi.org/10.1016/j.applanim.2007.06.001

van der Borg, J. A. M., Schilder, M. B. H., Vinke, C. M., & de Vries, H. (2015). Dominance in domestic dogs: A quantitative analysis of its behavioural measures. *PLOS ONE*, *10*, e0133978. https://doi.org/10.1371/journal.pone.0133978

van Dierendonck, M. C., de Vries, H., & Schilder, M. B. H. (1995). An analysis of dominance, its parameters and possible determinants in a herd of Icelandic horses. *Netherlands Journal of Zoology*, *45*, 362–385.

van Hooff, J. A. R. A. M., & Wensing, J. A. B. (1987). Dominance and its behavioural measures in a captive wolf pack. In H. Frank (Ed.), *Man and wolf* (pp. 219–252). Junk.

Vervaecke, H., de Vries, H., & van Elsacker, L. (2000). Dominance and its behavioral measures in a captive group of bonobos (*Pan paniscus*). *International Journal of Primatology*, *21*, 47–68. https://doi.org/10.

1023/A:1005471512788
Vervaecke, H., Stevens, J. M. G., Vandemoortele, H., Sigurjónsdóttir, H., & de Vries, H. (2007). Aggression and dominance in matched groups of subadult Icelandic horses (*Equus caballus*). *Journal of Ethology*, *25*, 239–248. https://doi.org/10.1007/s10164-006-0019-7
Vogel, E. R. (2004). *The ecological basis of aggression in white-faced capuchin monkeys,* Cebus capucinus*, in a Costa Rican dry forest* [PhD thesis]. Stony Brook University.
Watson, J. R. (1970). Dominance-subordination in caged groups of house sparrows. *The Wilson Bulletin*, *82*, 268–278.
Watt, D. J. (1982). Do birds use color bands in recognition of individuals? *Journal of Field Ornithology*, *53*, 177–179.
Watt, D. J. (1986). Relationship of plumage variability, size and sex to social dominance in Harris' sparrows. *Animal Behaviour*, *34*, 16–27. https://doi.org/10.1016/0003-3472(86)90002-3
Watt, D. J., & Sadowski, L. A. (1985). Development of courtship feeding in the American goldfinch (*Carduelis tristis*). *Proceedings of the Indiana Academy of Science*, *95*, 547–554.
Watts, C. (1969). *The social organization of wild turkeys on the Welder Wildlife Refuge, Texas* [PhD thesis]. Utah State University.
Watts, D. P. (1985). Relations between group-size and composition and feeding competition in mountain gorilla groups. *Animal Behaviour*, *33*, 72–85. https://doi.org/10.1016/S0003-3472(85)80121-4
Watts, D. P. (1994). Agonistic relationships between female mountain gorillas (*Gorilla gorilla beringei*). *Behavioral Ecology and Sociobiology*, *34*, 347–358. https://doi.org/10.1007/BF00197005
Watts, D. P. (1998). Coalitionary mate guarding by male chimpanzees at Ngogo, Kibale National Park, Uganda. *Behavioral Ecology and Sociobiology*, *44*, 43–55.
Weeks, J. W., Crowell-Davis, S. L., Caudle, A. B., & Heusner, G. L. (2000). Aggression and social spacing in light horse (*Equus caballus*) mares and foals. *Applied Animal Behaviour Science*, *68*, 319–337. https://doi.org/10.1016/S0168-1591(99)00126-4
Wells, S. M., & von Goldschmidt-Rothschild, B. (1979). Social behaviour and relationships in a herd of Camargue horses. *Zeitschrift Für Tierpsychologie*, *49*, 363–380. https://doi.org/10.1111/j.1439-0310.1979.tb00299.x
Wessling, E. G. (2011). *Rank-related differences in the travel patterns of savanna chimpanzees (*Pan troglodytes verus*) at Fongoli, Senegal* [Master's thesis]. Iowa State University.
Wheatley, B. P. (1978). *The behavior and ecology of the crab-eating macaque (*Macaca fascicularis*) in the Kutai Nature Reserve, East Kalimantan, Indonesia* [PhD thesis]. University of California.
Whitfield, D. P. (1985). *Social organisation and feeding behaviour of wintering turnstone (*Arenaria interpres*)* [PhD thesis]. University of Edinburgh.
Whitten, P. L. (1983). Diet and dominance among female vervet monkeys (*Cercopithecus aethiops*). *American Journal of Primatology*, *5*, 139–159. https://doi.org/10.1002/ajp.1350050205
Wickings, E. J., & Dixson, A. F. (1992). Testicular function, secondary sexual development, and social status in male mandrills (*Mandrillus sphinx*). *Physiology & Behavior*, *52*, 909–916. https://doi.org/10.1016/0031-9384(92)90370-H
Wiedenmann, R. N., & Rabenold, K. N. (1987). The effects of social dominance between two subspecies of dark-eyed juncos, *Junco hyemalis*. *Animal Behaviour*, *35*, 856–864. https://doi.org/10.1016/S0003-3472(87)80121-5
Wierenga, H. K. (1990). Social dominance in dairy cattle and the influences of housing and management. *Applied Animal Behaviour Science*, *27*, 201–229. https://doi.org/10.1016/0168-1591(90)90057-K
Williams, D. M. (1982). *Agonistic behaviour and mate selection in the mallard (*Anas platyrhynchos*)* [PhD thesis]. University of Leicester.
Williams, W. T., Kikkawa, J., & Morris, D. K. (1972). A numerical study of agonistic behaviour in the greybreasted silvereye (*Zosterops lateralis*). *Animal Behaviour*, *20*, 155–165. https://doi.org/10.1016/S0003-3472(72)80186-6
Wirtz, J. H. (1967). Social dominance in the golden-mantled ground squirrel, *Citellus lateralis chrysodeirus* (Merriam). *Zeitschrift Für Tierpsychologie*, *24*, 342–350. https://doi.org/10.1111/j.1439-0310.1967.tb00583.x
Wittig, R. M., & Boesch, C. (2003). Food competition and linear dominance hierarchy among female chimpanzees of the Taï National Park. *International Journal of Primatology*, *24*, 847–867. https://doi.

org/10.1023/A:1024632923180
Woolfenden, G. E., & Fitzpatrick, J. W. (1977). Dominance in the Florida scrub jay. *The Condor*, *79*, 1–12. https://doi.org/10.2307/1367524
Yamaguchi, T., Ito, M., Saeki, D., & Onishi, Y. (2008). Scaling social order in pigeons. *Japanese Journal of Animal Psychology*, *58*, 133–138. https://doi.org/10.2502/janip.58.2.1
Yasukawa, K., & Bick, E. I. (1983). Dominance hierarchies in dark-eyed juncos (*Junco hyemalis*): A test of a game-theory model. *Animal Behaviour*, *31*, 439–448. https://doi.org/10.1016/S0003-3472(83)80064-5
Zhang, S.-Q., Cui, Z.-W., Zhang, Y.-F., Wang, B.-S., Zhu, M.-L., Lu, J.-Q., & Wang, Z.-L. (2018). Low-ranking individuals present high and unstable fecal cortisol levels in provisioned free-ranging adult male rhesus macaques (*Macaca mulatta*) during the birth season in a mountain area of northern China. *Primates*, *59*, 517–522. https://doi.org/10.1007/s10329-018-0692-5
Zine, M. J., & Krausman, P. R. (2000). Behavior of captive mountain sheep in a Mojave Desert environment. *The Southwestern Naturalist*, *45*, 184–195. https://doi.org/10.2307/3672460
Zink, R. M., & Watt, D. J. (1987). Allozymic correlates of dominance rank in sparrows. *The Auk*, *104*, 1–10. https://doi.org/10.2307/4087226